

Small bites, big effects? Snacking and Health in the U.S.

Victoria Milani

September 29, 2026

Abstract

American children’s diets fall well short of federal dietary guidelines. According to the Healthy Eating Index (HEI), children score a mean of 47 out of 100 against a guideline benchmark of 80, and the score declines by roughly ten points from childhood to adolescence. Snacking accounts for about a quarter of children’s daily calories and is a plausible driver of this deficit, but its causal effect is confounded by selection on unobservable preferences over food and health. Using NHANES data from 2003 to 2018, I implement a bunching design at zero snack consumption to recover the marginal causal effect of habitual snack intake on the HEI and on the BMI z-score. An additional 100 kcal of habitual snacking lowers the HEI total score by 0.52 points. Removing a typical adolescent’s 300 kcal of daily snacking would raise the score by about 1.6 points, roughly 5 percent of the 33-point distance from the mean to the guideline benchmark. The effect decreases with age and is homogeneous across income levels. I find no BMI effect, rejecting z-score changes larger than 0.019 (about 0.2 kg of adolescent body weight) per 100 kcal with 95 percent confidence. These results suggest that what children snack on matters more for diet quality than how much they snack, so the margin with room in it is replacing unhealthy snacks rather than removing them.

1 Introduction

Human capital is built in early childhood, where initial investments are the most effective because they raise the return on every investment that follows (Cunha & Heckman, 2007; Heckman, 2007). Health is a core part of that early foundation, as good childhood health and nutrition lead to lasting benefits for education, earnings, and health later in life (Almond & Currie, 2011; Bleakley, 2007; H. Hoynes et al., 2016). Diet quality is the everyday channel through which that nutrition operates, and a diet that scores low on the Healthy Eating Index (HEI), the U.S. Department of Agriculture’s summary measure of how closely eating aligns with the federal Dietary Guidelines for Americans, predicts higher adult rates of cardiovascular disease, type 2 diabetes, and mortality regardless of body weight (Micha et al., 2017; Schwingshackl et al., 2018). Programs that raise the quality of what children eat carry measurable long-run returns. A Norwegian school breakfast rollout raised completed education and adult earnings (Bütikofer et al., 2018), and Californian schools that switched to a healthier meal vendor saw higher test scores without any change in obesity (Anderson et al., 2018).

American children eat badly, and snacking is a plausible reason why. They average a score of 47 out of 100 on the index, well below the federal benchmark of 80, and their scores drop another ten

points as they move into adolescence. This is particularly concerning because dietary habits built in childhood tend to carry into adulthood (Birch, 1999; Mikkilä et al., 2005). Because snack foods are far less nutrient-dense than the foods eaten at breakfast, lunch, and dinner, each additional snack calorie lowers overall diet quality. This effect is magnified by the fact that snacks now supply about a quarter of a child’s daily calories, up from 18 percent in the 1970s (Piernas & Popkin, 2010).

Changing children’s eating habits is difficult because snacking is established early and reinforced every day. Children are less forward looking than adults (Steinberg et al., 2009), and the food market capitalizes on that by designing snacks to be cheap, tasty, and heavily advertised. For busy parents, a quick snack is simply the path of least resistance, which means information campaigns alone rarely work. The policy instruments that have been studied causally act on structured eating occasions and leave the at-home snack margin to households. These include school meal programs (Hinrichs, 2010; Schanzenbach, 2009), beverage taxes (Allcott et al., 2019; Cawley & Frisvold, 2017), and SNAP benefits (H. W. Hoynes & Schanzenbach, 2009; Kreider et al., 2012). Existing pediatric guidance on snacking rests on cross-sectional comparisons of children who snack more or less (Hess et al., 2016), but snacking is endogenous to the household environment that produces it, so those comparisons cannot say what shifting the margin would deliver. This paper investigates which margins can be shifted and whether doing so delivers real value.

To address this, I estimate the marginal causal effect of an additional 100 kcal of habitual snack intake on children’s diet quality and body composition. Diet quality is captured by the total Healthy Eating Index score, and body composition is measured by age- and sex-standardized BMI z-scores. I use NHANES data from 2003 to 2018 and a bunching design (Caetano et al., 2024b) at zero snack consumption, where a snack is any eating or drinking occasion outside of breakfast, lunch, and dinner. The design exploits the mass of non-snackers at zero to recover the per-kcal causal effect at the bunching margin. That estimate is net of the household and behavioral characteristics that drive the raw cross-sectional correlation.

Isolating the causal effect requires separating the impact of snacking from selection into snacking, since unobserved household traits normally confound the two. The design starts from the fact that consumption cannot be negative. This creates a concentration of non-snackers at zero calories, accounting for 11.1 percent of the 25,220 children in the body-composition sample (21,508 in the diet-quality sample). Because non-snackers and the lightest snackers consume virtually the same amount, any difference in outcomes between them comes from selection rather than snacking itself. Two assumptions allow this local selection to identify the full causal effect. First, linear endogeneity assumes that selection operates at a constant rate across all levels of latent demand, so the selection measured near zero applies across all consumption levels. Second, tail symmetry assumes that latent demand among non-snackers mirrors the observed distribution of snackers around zero, allowing the data to determine the composition of the zero group. Subtracting the implied selection from the raw relationship then leaves the causal effect of a marginal snack calorie.

Snacking lowers overall diet quality without altering body composition. Every additional 100 calories of daily snack intake reduces the HEI score by 0.52 points. This reduction is strongest

among children ages 2 to 5 at 1.12 points. It decreases to 0.42 points among adolescents and holds uniform across income levels at about 0.5 points. In contrast, the estimated effect on BMI z-scores is a precise zero at 0.005 per 100 calories, with a 95% confidence interval of $[-0.009, 0.019]$. Even at its upper bound, this amounts to just 0.2 kilograms of body weight for an adolescent. Because this null finding holds across age groups, income levels, and alternative snack definitions, the impact of snacking clearly operates through nutrition rather than weight.

To put this magnitude into perspective, consider a typical adolescent who habitually consumes 300 snack calories per day, equivalent to a small bag of chips and a fruit-flavored drink. Removing that intake raises the child’s diet score by about 1.6 points, which covers roughly 5% of the 33-point gap between the population mean of 47 and the federal benchmark of 80. For context, major national interventions like the Healthy, Hunger-Free Kids Act school meal reforms raised participants’ whole-day score by 1.9 to 4.7 points (Kinderknecht et al., 2020). Reducing a typical adolescent’s daily snacking moves diet quality by an amount comparable to a major federal reform of school food.

The drop in diet quality is concentrated heavily in sweet snacks. Every 100 calories of sweet snacks reduces the diet score by 0.85 points, compared to a 0.63 point drop for savory snacks, making sweet snacks about 35% more harmful per calorie. This difference comes down to ingredients. Sweet snacks are filled with added sugars and refined grains that trigger specific penalties on the HEI at the levels children typically eat them. For example, eliminating a 300 calorie sweet snack combination, like a chocolate bar and a small juice box, raises diet quality by roughly 2.6 points, rather than the pooled 1.6.

High-glycemic snacks carry an even larger penalty, reducing the diet score by 1.49 points per 100 calories against 0.74 for snacks in general, both measured under the same alternative estimator.¹ Removing 300 kcal of high-GI snacks raises the HEI by about 4.5 points. Furthermore, snacks add to mealtime eating rather than replacing it. An additional 100 calories of snacks increases non-snack consumption by 27 calories, turning a 100 calorie snack into a 127 calorie net increase in daily intake. Despite this total caloric addition, the overall effect on BMI remains a precise zero, which is consistent with normal childhood growth and energy expenditure absorbing the extra calories at the population level.

What children snack on therefore matters more than how much they snack. Because the per-calorie penalty depends so strongly on which foods supply the calories, changing the contents of a snack should do more than shrinking it. Holding each child’s snack calories fixed and reassigning them to snack foods these same children already eat lifts the average whole-day score by 9.7 points, roughly six times the 1.6 points available from removing 300 kcal. That number assumes full compliance and no adjustment elsewhere, so it bounds the available gain rather than predicting it, and it carries no weight implication because the calories never leave the day.

These findings contribute directly to the economic literature on childhood obesity and food

¹The tail-symmetric estimator used for the headline results is not well identified for the high-GI subset, because 83% of children consume no high-GI snacks on a given day, a bunching share above the one-half ceiling for the tail-symmetric construction. The semiparametric-normal specification is reported instead, so the comparison is like-for-like against the pooled semiparametric-normal estimate of -0.74 rather than the tail-symmetric -0.52 .

choice, which seeks to isolate the causal impacts of the food environment from unobserved household factors (Cawley, 2010; Cawley & Meyerhoefer, 2012). Prior research in this area largely focuses on institutional policies that shift diet without altering underlying family traits. These include school meal programs (Hinrichs, 2010; Schanzenbach, 2009), beverage and sugar taxes (Allcott et al., 2019; Cawley & Frisvold, 2017), the Supplemental Nutrition Assistance Program and similar nutrition assistance schemes (H. W. Hoynes & Schanzenbach, 2009; Kreider et al., 2012), and changes in local food access and prices (Currie et al., 2010; Handbury et al., 2015). I extend that literature by measuring the causal effect of everyday eating outside structured meals, an unaddressed behavior identified here through a corner solution feature of the data rather than an institutional policy change.

Additionally, these findings contribute to the economics of early childhood and the human capital production function. In this framework, child body composition and diet quality function as both outputs of household investment and inputs to adult human capital (Almond & Currie, 2011; Bleakley, 2007; Cunha & Heckman, 2007; Heckman, 2007; H. Hoynes et al., 2016). This literature has had no clean causal evidence on habitual snack consumption, because everyday eating decisions are difficult to observe at scale and standard identification strategies do not apply. I provide a population-scale causal estimate for this specific dietary behavior.

Methodologically, the paper belongs to the bunching literature in applied microeconomics. Originally developed to study how individuals respond to taxes and government transfers, bunching designs traditionally rely on policy kinks or notches (Kleven, 2016; Saez, 2010). This study uses a more recent corner solution variation where bunching occurs because intake cannot drop below zero, rather than from a policy rule. This framework was introduced in Caetano et al. (2024b) and has been applied to maternal labor supply (Caetano et al., forthcoming), children’s time use (Caetano et al., 2026), enrichment activity choices (Caetano et al., 2024a), and media consumption (Caetano et al., 2025). In each of those settings the corner is a choice not to participate, therefore, here it is a biological floor on consumption, which widens the design’s reach to behaviours no institution assigns and no instrument shifts.

The rest of the paper follows a straightforward structure. Section 2 describes the data, while Section 3 sets out the empirical strategy. Section 4 presents the main and heterogeneity estimates. Section 5 reports the robustness checks, and Section 6 concludes.

2 Data

The analysis draws on repeated cross-sectional data on children’s dietary intake, anthropometrics, and demographics from the National Health and Nutrition Examination Survey (NHANES), covering the eight two-year cycles from 2003–2004 through 2017–2018. The BMI sample contains 25,220 children aged 2 to 19, while the HEI sample contains 21,508 children, since it excludes the 2003–2004 cycle because the U.S. dietary guidelines changed in 2005. Section 2.1 describes the treatment variable, Section 2.2 the outcome variables, Section 2.3 the control variables and sample composition,

and Section 2.4 characterizes how much children snack, how that changes with age, and what fills the snack piece.

2.1 Treatment Variable

Following the nutritional literature (Hess et al., 2016), a snack is defined as any food or caloric beverage consumed outside of breakfast, lunch, and dinner. NHANES records 24 hour dietary recall data for each item together with a participant-supplied label for the eating occasion and the reported time of consumption, making it straightforward to isolate snack items and aggregate their total calories. Beverages consumed within thirty minutes of a main meal are assigned to that meal rather than flagged as snacks, while plain or flavored water is dropped because it contributes no calories. The treatment variable, denoted S_i for child i , is total caloric intake from snacks in the 24 hours prior to the interview.

Figure 1 summarizes the distribution of S_i in the sample. About 11.1% of children report no snack consumption on the previous day, generating a mass point at zero that motivates the empirical strategy of Section 3. Among children who do snack, caloric intake is right-skewed, averaging 563 kcal and a median of 435 kcal. The right panel of Figure 1 focuses on the interval $0 < S_i \leq 150$ calories, showing meaningful density even at very small positive amounts.

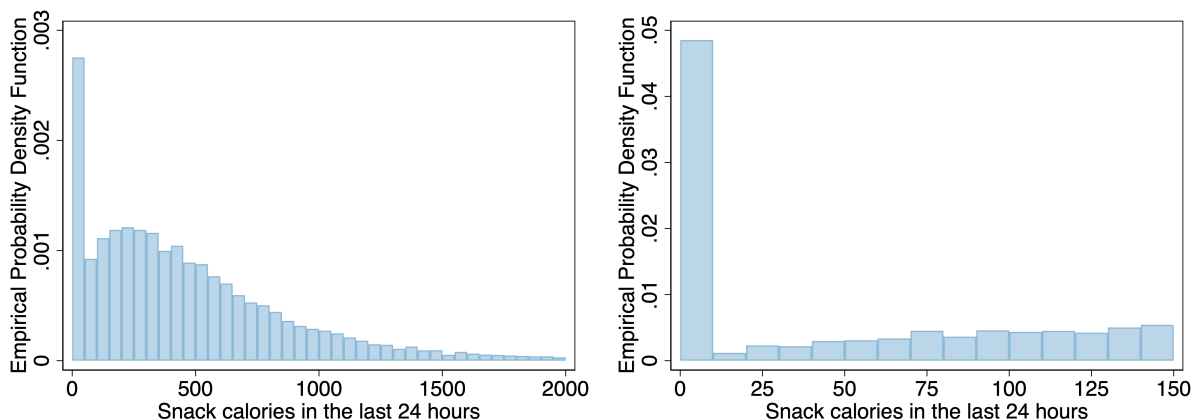


Figure 1: Distribution of snack caloric intake among snackers.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018, author’s calculations.

Notes: The left panel shows the density of S_i for children with $S_i > 0$. The right panel zooms in on $0 < S_i \leq 150$ kcal to display the mass at very low positive intakes.

2.2 Outcome Variables

I study two outcomes, physical growth and the Healthy Eating Index (HEI), which capture what dietary intake does to a child’s body and to the quality of the diet itself. Growth is measured by body mass index (BMI) and, in Section 4.2, by height and weight separately. The separation matters because BMI is a ratio, W/H^2 , and a child whose mass and stature both rise in proportion

registers no change in it. Reading BMI as adiposity alone therefore discards the margin along which children actually respond to energy. Attained height in childhood is a standard measure of net nutrition in economics and predicts adult earnings, cognitive performance, and longevity (Deaton, 2007; Steckel, 2009), while the HEI predicts adult chronic-disease incidence and all-cause mortality in prospective cohorts (Micha et al., 2017; Schwingshackl et al., 2018). For the main pooled analysis, I use the age- and sex-standardized BMI z-score from the CDC growth reference, denoted $\text{BMI-}z_i$, so that estimates can be aggregated across the full 2-19 age range without mechanical scale differences across ages. The alternative-definition robustness table in Section 5 additionally reports raw BMI within age bins and income terciles, where the standardization is not needed and the raw scale keeps the estimates comparable across bins.

The HEI is a 100-point summary index of diet quality developed by the USDA, constructed from thirteen components that score adherence to dietary guidelines on dimensions such as total fruit, whole grains, dairy, total protein foods, sodium, and added sugars (National Cancer Institute, Division of Cancer Control and Population Sciences, 2024). A higher score indicates a diet more closely aligned with federal dietary recommendations. Across the HEI sample, children’s full-day diets average a total score of about 47 out of 100, against a federal guideline benchmark of 80, and the deficit is spread unevenly across the thirteen components. Table B.1 in Appendix B lists each of the thirteen components, the corresponding USDA dietary recommendation, and the scoring rule for each component. Because the Dietary Guidelines for Americans (DGA) changed from the 2000 to the 2005 standards, the HEI sample begins in 2005–2006. The BMI sample is unaffected by this restriction.

The HEI sample has a steep age profile and almost no income profile. Across single years of age, the mean HEI total score declines from 54.4 at age 2 to 44.0 at age 19, the ten-point childhood-to-adolescence drop referenced in the introduction (Figure 2, left panel). Income barely moves the score. Across deciles of the household poverty income ratio, the bottom decile averages 46.4, the middle deciles hover around 46.5–47.3, and the top decile reaches 48.9, an overall span of about 2.5 points (Figure 2, right panel). The diet-quality deficit is therefore tied to what children eat as they age rather than to what households can afford, consistent with a behavioral margin such as snacking rather than purchasing power driving it.

The total-score trends in Figure 2 hide which components of the HEI move with age and with income. Figure C.1 in Appendix C reports all 13 HEI-2015 component scores by age group and by income tercile, and the two gradients look nothing alike. The age decline in the HEI total is driven by sharp drops on fruits, dairy, added sugars, sodium, and refined grains across childhood, with total protein foods and fatty acids partially offsetting. The income gradient is much flatter. Most components shift by less than two points across terciles, and the substantive components that drive the age decline barely move with income. This is why the income line plot in Figure 2 appears flatter than the age line plot even though both summarize the same underlying index.

That the deficit tracks age rather than income points to a behavioral margin in what children eat, and snacking is the natural place to look. To characterize that margin, Table B.2 in Appendix B

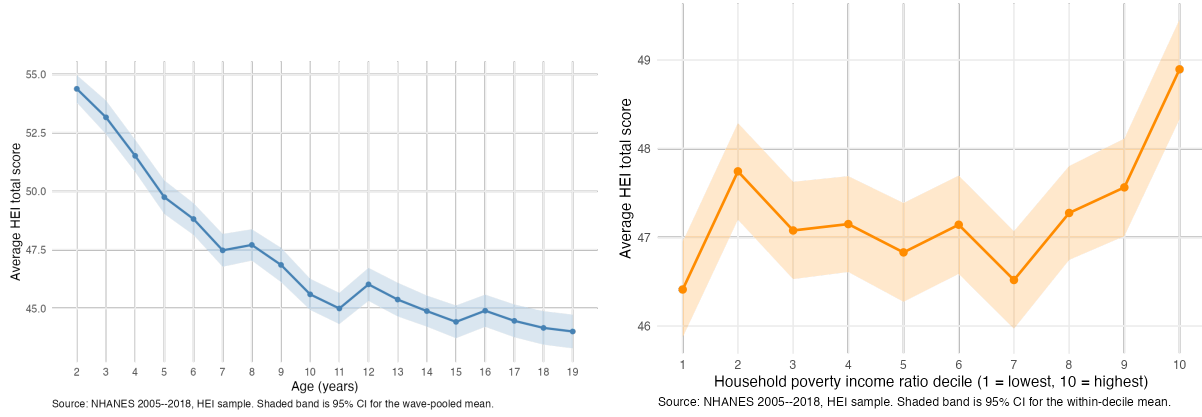


Figure 2: Mean HEI total score by single year of age (left panel) and by decile of the household poverty income ratio (right panel).

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, HEI sample, author’s calculations.

Notes: Shaded bands are 95% confidence intervals.

reports, for each simplified food category, the share of snacking children who consume it and the average kcal they consume when they do. The dedicated “Snacks” category is the most common, reaching 79% of snacking children at about 325 kcal per day among those who eat it, followed by beverages (69% of snackers, 111 kcal). Fruits and vegetables reach a little over a third of snackers (39%, 128 kcal) and dairy and eggs about a third (32%, 180 kcal). A handful of categories are consumed rarely but in large amounts when they are, most notably mixed dishes (10% of snackers but 403 kcal), so prevalence and intensity capture different parts of the margin.

Both outcomes display a visible jump at the bunching point $S_i = 0$ before any controls are applied. Figure 3 in the next subsection reports the local mean of the BMI z-score and the HEI total score against S_i in a neighborhood of zero, together with the analogous local means for the observable controls in X_i . The bunched group sits noticeably below the snacker-side limit for BMI z-score and noticeably above for HEI. Both jumps will form the basis of the empirical strategy in Section 3, where the design isolates how much of each gap reflects the causal effect of snacking and how much reflects systematic differences between bunched and non-bunched children.

2.3 Demographics and Sample Composition

Age in years and its square, sex, race, the child’s educational attainment, household poverty income ratio and income tercile, and food security status have each been linked by prior work to children’s dietary intake or to BMI. Together with NHANES cycle fixed effects, which absorb secular shifts in dietary intake and survey methodology across the 2003–2018 period, these covariates form the vector X_i that the empirical strategy conditions on. To avoid losing observations to item nonresponse in any of them, I apply the missing-indicator method, recoding missing values of each covariate to a constant (zero for continuous variables, a separate “missing” category for categorical ones) and

adding to X_i a binary indicator for whether the original value was missing. This preserves the full analytic sample and absorbs any systematic difference associated with missingness. The estimates are insensitive to this convention. A variant that recodes missing continuous controls to the sample mean rather than to zero is included in the specification curve discussed in Section 5.

Table 1 reports summary statistics for the BMI and HEI samples. Non-snackers are on average younger and are over-represented in food-insecure and lower-income households, so the children who bunch at zero are not a random subset of the population. They differ from children with arbitrarily low positive S_i along these observables and, absent an implausible independence assumption, along unobservables correlated with BMI and diet quality. This is the selection problem the bunching design of Section 3 corrects.

Figure 3 plots, alongside the two headline outcomes in the top row, the local mean of four covariates in X_i as a function of S_i near zero, the household poverty income ratio and food security in the middle row and white race and age below. Each covariate jumps visibly at $S_i = 0$. The compositional difference the table documents on average therefore persists at the margin, between the bunched children and those who barely snack. The remaining controls, and the child’s dieting status, show the same pattern in Appendix Figure C.2. A marginal change in snacking cannot move a child’s household income, food security, or race, so these jumps can only reflect a difference in who bunches at zero, not an effect of snacking. They are why the raw outcome gap at $S_i = 0$ cannot be read as a causal effect, and why the design of Section 3 is needed to separate snacking from selection.

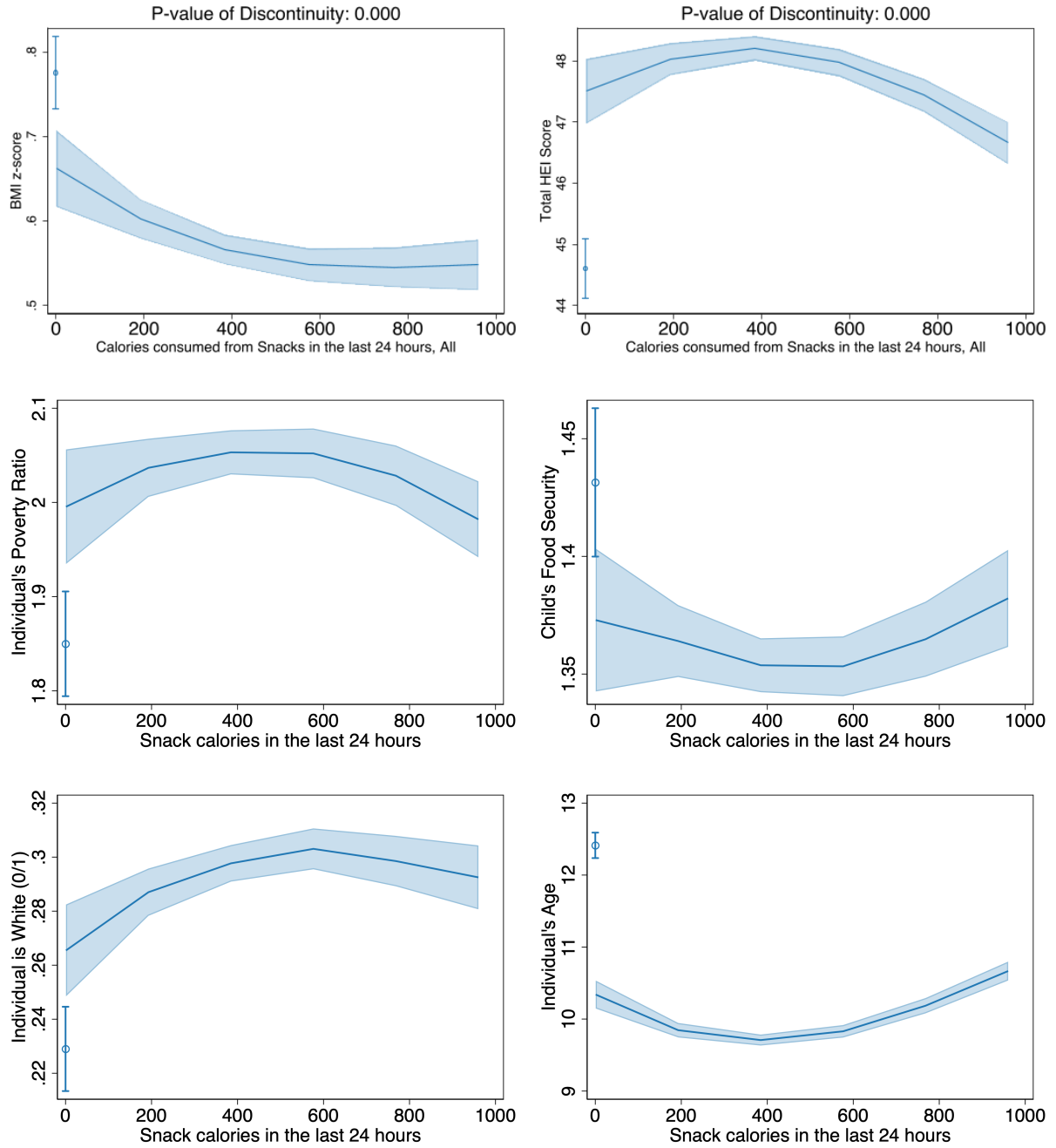


Figure 3: Local mean of both headline outcomes and selected demographic covariates as a function of snack caloric intake S_i in a neighborhood of zero.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI panel) and 2005–2018 (HEI panel), author's calculations.

Notes: Top row: BMI z-score (left) and HEI total score (right). Middle row: household poverty income ratio (left) and an indicator for food security (right). Bottom row: an indicator for white race (left) and age in years (right). In every panel the point at $S_i = 0$ is the mean among bunched children with a 95% confidence interval, and the line on the positive side is a degree-1 local polynomial smoother on snackers with a bandwidth of 250 kcal. The remaining controls in X_i , together with the child's dieting status, are shown in Appendix Figure C.2.

Table 1: Summary statistics by sample and snacking status.

	BMI sample			HEI sample		
	Non-snackers	Snackers	All	Non-snackers	Snackers	All
<i>Outcomes</i>						
BMI (kg/m ²)	22.65 (6.82)	20.40 (5.76)	20.65 (5.93)	–	–	–
BMI z-score (CDC reference)	0.78 (1.17)	0.57 (1.17)	0.59 (1.17)	–	–	–
HEI total score	–	–	–	44.61 (12.19)	47.68 (12.53)	47.33 (12.53)
<i>Treatment and daily intake</i>						
Snack calories (kcal)	0.00 (0.00)	562.71 (504.54)	500.55 (507.49)	0.00 (0.00)	547.32 (490.00)	485.33 (492.94)
Total daily calories (kcal)	1701.36 (825.39)	1979.60 (890.77)	1948.86 (888.06)	1674.09 (813.22)	1948.75 (871.60)	1917.64 (869.54)
<i>Demographics</i>						
Age (years)	12.41 (4.76)	10.11 (5.24)	10.37 (5.23)	12.30 (4.78)	9.92 (5.20)	10.19 (5.21)
Female (%)	46.4	49.6	49.3	47.2	49.5	49.2
Non-Hispanic white (%)	22.9	29.5	28.7	23.4	29.6	28.9
Household poverty ratio	1.85 (1.43)	2.03 (1.54)	2.01 (1.53)	1.85 (1.44)	2.05 (1.54)	2.02 (1.53)
<i>Health and behavior</i>						
Diagnosed diabetes (%)	1.1	0.7	0.8	1.1	0.7	0.8
On a special diet (%)	7.0	4.3	4.6	6.9	4.4	4.6
Self-rated diet quality (1=excellent, 5=poor)	3.08 (0.99)	3.09 (1.00)	3.08 (1.00)	3.08 (0.99)	3.09 (1.00)	3.08 (1.00)
Any food insecurity (%)	45.3	39.8	40.4	46.0	40.5	41.1
Observations	2,786	22,434	25,220	2,436	19,072	21,508

2.4 Patterns of Children’s Snacking

Before turning to identification, this subsection characterizes the behavior the paper estimates the effect of. Three features matter for what follows. Snacking is close to universal and accounts for a quarter of children’s energy, its caloric level grows with age while its budget share does not, and the foods that fill it differ systematically from the foods children eat at meals.

Prevalence and scale. Snacking is the modal way American children eat outside of meals. In the HEI sample, 88.7% of children record at least one snack occasion with positive energy on their recall day, leaving 11.3% at the corner that identifies the design. Among all children, snack occasions supply 485 kcal per day on average, or 25.3% of the 1,918 kcal daily total. That share is large enough that any change in what fills it moves whole-day diet quality mechanically, before any behavioral response in the rest of the day.

Age profile. The caloric level of snacking rises steadily across childhood while its share of the diet does not. Children aged 2 to 5 take 444 kcal from snacks against a 1,535 kcal daily total, children aged 6 to 11 take 468 kcal against 1,921, and adolescents aged 12 to 19 take 522 kcal against 2,132. The budget share therefore falls from 28.9% in early childhood to 24.4% and 24.5% in the two older bins, so snacking grows in absolute terms roughly in proportion to total intake. The corner itself thins with age in the opposite direction. The fraction of children reporting no snack calories rises from 5.4% at ages 2 to 5 to 9.2% at 6 to 11 and 16.4% at 12 to 19, which means the bunching mass that identifies the estimator is three times larger among adolescents than among preschoolers. Figure ?? in Appendix C reports the share and the level side by side.

What children snack on. [TO COMPUTE] Table ?? reports the composition of snack calories along the four margins the paper later exploits. Sweet snacks account for XX% of snack energy and savory snacks for XX%. High-glycemic foods, defined by a glycemic index of 70 or above, account for XX%. Sugar-sweetened beverages contribute XX kcal per day on their own, and liquid calories more broadly make up XX% of the snack piece against XX% of the meal piece. These splits are constructed from `sweet_snack_kcal`, `savory_snack_kcal`, `high_gi_snack_kcal`, `ssb_snack_kcal`, and `liquid_snack_kcal`, all of which are already on the estimation file.

Snack foods against meal foods. The nutritional distinctness of the snack piece is what makes the composition channel plausible. Comparing the snack piece and the meal piece of the same child on the same day, per 1,000 kcal, snack calories carry 30 to 50% less protein, iron, zinc, folate, and B12, and roughly twice the added sugar. Because the comparison holds the child fixed, it cannot be explained by differences between children who snack more and less. Figure ?? in Appendix C reports the full nutrient panel.

When children snack. [TO COMPUTE, REQUIRES DATA BUILD] The timing of snack occasions is not yet on the estimation file. NHANES records the clock time of every eating occasion in DR1_020, which would support a distribution of snack energy across the day, the share falling after the evening meal, and whether the late-evening portion is disproportionately sweet or high-glycemic. The NoNight and PM estimation samples already in Data_Modified suggest part of this split exists upstream and could be surfaced here. Timing matters substantively because the insulin-response channel developed in Section 4.2 depends on when energy arrives relative to sleep, not only on how much arrives.

3 Empirical Strategy

The empirical strategy identifies the causal effect of snacking on BMI and diet quality from the bunching of children at zero snack consumption. I follow Caetano et al. (2024b), Caetano et al. (forthcoming), applying the framework to a setting where the bunching point arises from a non-negativity constraint on observed intake. Section 3.1 explains why standard observational approaches are infeasible here, Section 3.2 develops the intuition for how bunching at zero carries information about the unobserved confounders, Section 3.3 states the formal identification result and the assumptions behind it, and Section 3.4 describes the estimator and inference procedure.

3.1 Why Standard Approaches Are Infeasible

The simplest way to estimate the effect of snacking on a child health outcome is to regress the outcome on snack caloric intake with controls for observable characteristics. However, that regression is not credible here, because snack consumption is endogenous to dietary preferences and unobserved household resources that affect health directly. A child who snacks frequently likely comes from a household with a particular pattern of food availability, parental time, and dietary norms, and any of these can move BMI or diet quality through channels unrelated to snacking. Controls absorb part of this confounding, substantial bias remains because of the persistent sociodemographic patterning in child food environments described in Section 2.3.

The conventional fixes for endogeneity are also hard to apply in this setting. An instrument that shifts snack consumption without touching health through other channels is not readily available, and school-based or geographic variation in snack availability typically operates through other dietary margins as well.

First-differencing across the two NHANES 24-hour dietary recalls is a candidate, since it would eliminate any time-invariant individual traits such as baseline food preferences, household dietary norms, and metabolic predisposition from the estimating equation. However, this approach fails for two reasons. First, day to day changes in a child’s snacking are driven by temporary unobserved events like a busy family schedule, a school holiday, or a sudden spike in hunger, so differencing trades a time-invariant confounder for a time-varying one. Second, NHANES measures body mass index only once during the physical exam, leaving no within-child variation for weight outcomes,

dropping its headline body-composition outcome from scope.

The bunching design exploits the one feature of the problem that these approaches leave untouched, the non-negativity constraint on snack consumption itself. Because $S_i \geq 0$ mechanically, children whose unconstrained S_i^* would be zero or negative pool at the bunching point $S_i = 0$. The size and composition of this mass point, combined with a linear specification of the selection function and a distributional assumption on the tail of S_i^* , identify the average causal effect of S_i^* at the bunching margin. This framework recovers causal effects without requiring external instruments or institutional policy thresholds.

The design does not claim that snack consumption is exogenous. Instead, exogeneity stems from the biological non-negativity constraint, which piles up all children with zero or negative underlying propensities to snack at zero. The structure of that mass point reveals how the unobservable confounders shift the outcome at the corner. By subtracting this baseline shift from the relationship observed among snack consumers, the design isolates the per calorie causal effect of latent snack demand from individual selection.

3.2 Intuition of the Identification Approach

Non snackers on a given recall day represent a diverse group with varying underlying propensities to eat. Some children barely refrain and would snack with minimal encouragement, whereas others hold deeply negative latent demand and require a substantial push to eat snacks. A naive comparison between snackers and non snackers treat this bunched group as homogeneous, masking the unobserved variation inside zero consumption that generates selection bias.

Let S_i denote the observed snack caloric intake of child i on the recall day, and let S_i^* denote the latent index capturing the child’s underlying propensity to consume snack calories, encompassing all the factors correlated with S_i . The mapping between observed consumption and latent demand follows the corner-solution structure

$$S_i = S_i^* \cdot \mathbf{1}(S_i^* \geq 0), \quad \Pr(S_i^* < 0) > 0. \quad (1)$$

where $\mathbf{1}(\cdot)$ is an indicator function equal to one when $S_i^* \geq 0$ and zero otherwise.

Children with positive latent demand consume the amount their preferences and constraints dictate, so $S_i = S_i^*$ whenever $S_i > 0$. In contrast, children with non-positive latent demand pool at $S_i = 0$, regardless of how negative their S_i^* may be. The index aggregates all factors correlated with snack consumption into a single dimension, including child preferences, household food access, parental rules, school policies, and food security constraints. Consequently, Equation (1) serves as a definitional identity rather than a behavioral model, making no causal claim about S_i^* itself.

Figure 4 plots the cumulative distribution of S_i for the pooled sample and separately for the three age groups used in the heterogeneity analysis. The vertical jump at zero in each panel represents the share of children who report no snacking on the recall day. This zero-intake share rises with age, ranging from approximately 7% in the youngest group to about 20% in the oldest. Above zero, the cumulative distribution is smooth across all age categories, confirming that snack intake among

consumers is continuous. The estimation design exploits only this bunching at the zero lower bound.

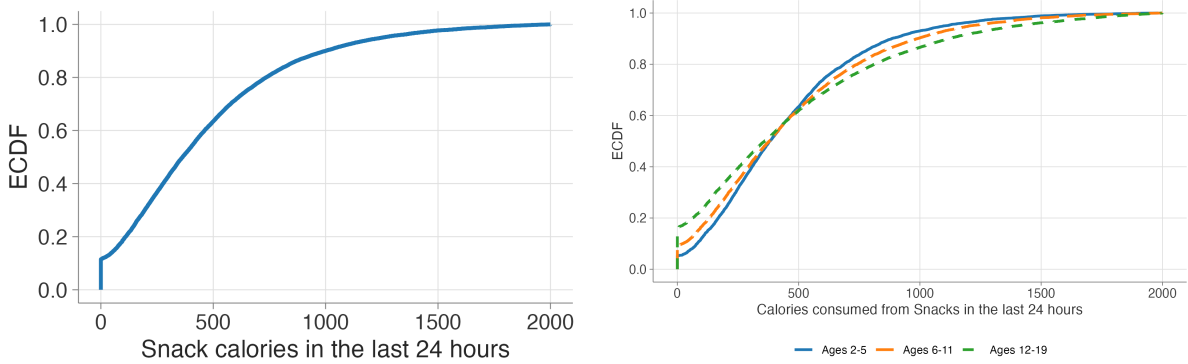


Figure 4: Empirical cumulative distribution function of snack caloric intake S_i .

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018, author’s calculations.

Notes: The left panel pools all children. The right panel splits the sample by age bin. The vertical jump at $S_i = 0$ in each panel is the share of children with no reported snack consumption.

This bunching matters because the group of children at $S_i = 0$ is internally heterogeneous in S_i^* . It mixes children at the margin of indifference ($S_i^* = 0$) with those whose preferences and constraints place the strictly below zero ($S_i^* < 0$). Consequently, the average latent demand among non-snackers, $\mathbb{E}[S_i^* | S_i = 0]$, is strictly negative. In contrast, because $S_i^* = S_i = s$ for all positive intake $s > 0$, average latent demand converges to zero as consumption shrinks toward zero. Formally, this creates a sharp jump in conditional expectation at the boundary:

$$\lim_{s \downarrow 0} \mathbb{E}[S_i^* | S_i = s] - \mathbb{E}[S_i^* | S_i = 0] = -\mathbb{E}[S_i^* | S_i = 0] > 0, \quad (2)$$

This discontinuity at $S_i = 0$ measures the exact magnitude of average latent demand among the bunched group.

Equation (2) alone is not sufficient to identify the causal effect for two reasons. First, the relationship between S_i^* and the health outcome is itself confounded by unobserved factors. Second, the magnitude of the discontinuity depends on $\mathbb{E}[S_i^* | S_i = 0]$, an expectation over an unobservable variable.

Section 2.3 provides indirect empirical evidence for the discontinuity in Equation (2). Observable variables correlated with S_i^* are themselves discontinuous at $S_i = 0$, with non-snackers differing noticeably from children with trace positive intake across household income, food security, and race. Differences of this magnitude cannot plausibly be the treatment effect of consuming a negligible amount of snack calories. Instead, they reflect a discontinuity in the underlying S_i^* at the zero boundary.

Because the health outcome depends on S_i^* through unobservable confounders, this same discontinuity appears in the conditional mean of the outcome at $S_i = 0$. Figure 5 plots the conditional mean of each outcome in a neighborhood of $S_i = 0$ after netting out the observable controls X_i

described in Section 2.3, with the BMI z-score in the left panel and the HEI total score in the right.

To construct these plots, I first fit, on snackers only ($S_i > 0$), the regression

$$Y_i = a + bS_i + c'X_i + u_i,$$

and then construct the adjusted outcome $\tilde{Y}_i = Y_i - \hat{a} - \hat{c}'X_i = \hat{u}_i + \hat{b}S_i$ for the full sample, which removes the fitted contribution of the controls while preserving the relationship with S_i on the positive side.

The figure plots the mean of $\tilde{Y}_i$ at the bunching point together with a degree-1 local polynomial smoother of $\tilde{Y}_i$ fit on the positive side with a bandwidth of 250 kcal. Unlike the raw discontinuity plots of Figure 3, where the jump at zero conflates the outcome gap with the observable compositional gap between bunched and marginal children, the local means here are adjusted for observables. The jump that remains is the part of the outcome gap the observables do not explain, the part the design must decompose into the causal effect and the residual unobservable selection.

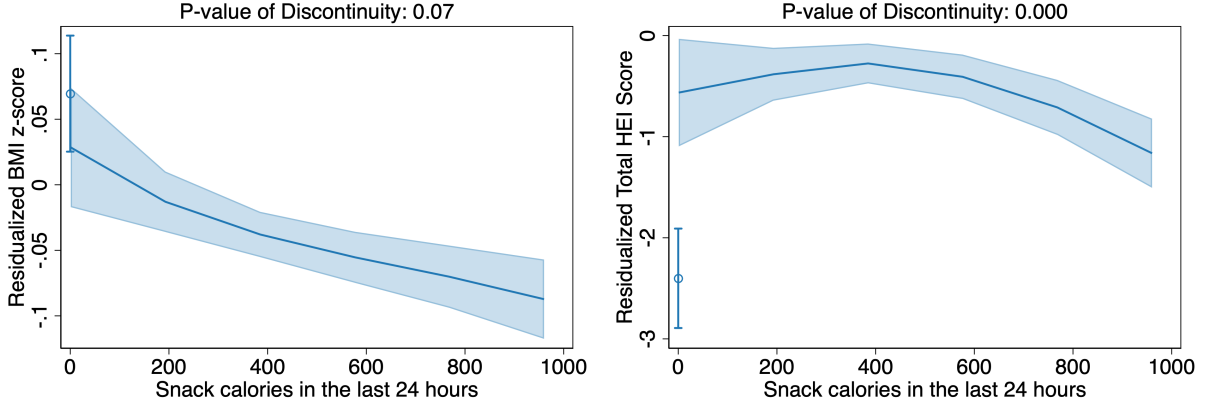


Figure 5: Residualized outcomes as a function of snack caloric intake in a neighborhood of zero.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI panel) and 2005–2018 (HEI panel), author’s calculations.

Notes: The left panel uses the BMI z-score, and the right panel uses the HEI total score. Both outcomes are residualized against the control vector X_i described in Section 2.3. The line on the positive side is a degree-1 local polynomial smoother with a bandwidth of 250 kcal.

Part of the vertical gap at $S_i = 0$ in each panel is the causal effect of moving from the value of S_i^* prevailing among bunched children to a small positive intake, and this part scales with the magnitude of the latent-demand discontinuity in equation (2), $-\mathbb{E}[S_i^* \mid S_i = 0]$, the quantity on which the identification hinges. The remainder is the difference in unobservable determinants of the outcome between bunched and non-bunched children at the margin. Neither component is observed directly, and the same observed gap is consistent with very different combinations of the two. The identification assumptions of the next subsection impose enough structure to separate them.

3.3 Formal Identification

Let Y_i denote the outcome of interest, either the BMI z-score for the body-composition analysis or the HEI total score for the diet-quality analysis. The outcome equation is

$$Y_i = \beta S_i + U_i, \quad (3)$$

where β is the per-kcal effect of interest and U_i absorbs all other determinants of the outcome, including a constant. Concerns about endogeneity in equation (3) are encapsulated in how U_i varies with S_i^* , since S_i^* is by construction a sufficient index of all confounders. Identification begins by restricting this dependence.

Assumption 1 (Linear endogeneity). $\mathbb{E}[U_i | S_i^*] = m_0 + \delta S_i^*$.

In words, Assumption 1 says that the conditional mean of the unobservable outcome determinant moves linearly with the latent snack propensity. The unobserved factors that lead a child to snack more, whether the household food environment, parental dietary norms, or the child’s own appetite, must operate in the same direction throughout the support of S_i^* , even if their magnitude varies.

When would the conditional mean bend rather than run straight? If it did, the single selection slope δ recovered on the positive side would not extrapolate to the bunched group, and the correction would be biased. One such pattern arises if health-conscious parents push their children toward fruit and other high-quality snacks while less health-conscious parents allow low-quality snacks, so that high snack intake is reached by two opposing types. This threatens the assumption only to the extent that it makes $\mathbb{E}[U_i | S_i^*]$ non-monotone in S_i^* , since the assumption restricts the average of U_i at each level of S_i^* rather than the behavior of any single family. The design does not require these parental types to behave alike. Its correction removes the selection component of the discontinuity, extrapolated from the snacking side under Assumption 1, and reads only the residual as the causal effect of snacking. Differences across types are therefore admissible as long as they account for part of the jump at $S_i = 0$ rather than the whole of it through a selection pattern the single linear slope cannot reproduce. The concern is further limited because the controls X_i absorb the observable part of parental health-consciousness, such as income, education, and food security, so the assumption need only hold within groups of observationally similar children. Section 5 moreover tests directly whether the conditional mean is approximately linear over the observed range.

Substituting Assumption 1 into equation (3) and rearranging gives

$$Y_i = m_0 + \beta S_i + \delta S_i^* + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | S_i^*] = 0, \quad (4)$$

so β and δ would be identified by an ordinary regression of Y_i on a constant, S_i , and S_i^* if S_i^* were observable. It is not, and the bunching design constructs a control function that substitutes for it. Using the identity $S_i^* = S_i + S_i^* \cdot \mathbf{1}(S_i = 0)$, which holds because $S_i^* = S_i$ on the positive side, and taking conditional expectations of equation (4) given S_i ,

$$\mathbb{E}[Y_i | S_i] = m_0 + (\beta + \delta) S_i + \delta \mathbb{E}[S_i^* | S_i = 0] \cdot \mathbf{1}(S_i = 0). \quad (5)$$

Equation (5) is the empirical counterpart of the residualized-outcome plot in Figure 5. The slope on the positive side identifies $\beta + \delta$, the sum of the causal effect and the selection effect, and the discontinuity at zero is $-\delta \cdot \mathbb{E}[S_i^* | S_i = 0]$. Both objects are observed, so knowing the magnitude of $\mathbb{E}[S_i^* | S_i = 0]$ would deliver δ from the ratio of the outcome discontinuity to the latent discontinuity and β by subtraction.

The signs of the two outcome discontinuities in Figure 5, combined with the negative value of $\mathbb{E}[S_i^* | S_i = 0]$ implied by equation (2), are internally consistent with this decomposition. For HEI, the residualized outcome at the bunching point sits below the snacker-side limit, which together with $\mathbb{E}[S_i^* | S_i = 0] < 0$ implies $\hat{\delta} > 0$. Snackers have unobservable characteristics that are independently associated with higher HEI, the positive selection on unobservables documented in Section 4, so the naive cross-sectional comparison understates the harm of snacking, and the correction subtracts this positive selection slope to leave the headline negative causal effect on diet quality. For the BMI z-score the residualized outcome at the bunching point sits above the snacker-side limit, the implied $\hat{\delta}$ is negative, and the correction subtracts a negative selection slope, leaving the positive-signed point estimate on body composition that Section 4 reads as a tightly identified zero.

Pinning down $\mathbb{E}[S_i^* | S_i = 0]$ requires knowing the shape of the lower tail of S_i^* , which is never observed. I recover it by reflecting the observed upper tail of the positive-side distribution across zero.

Assumption 2 (Tail symmetry). *The lower tail of the distribution of S_i^* below the $\Pr(S_i = 0)$ -th quantile is the mirror image of the upper tail above the $(1 - \Pr(S_i = 0))$ -th quantile.*

Since $S_i = S_i^*$ on the positive side, the upper tail of S_i^* is observed directly. Assumption 2 therefore identifies the lower tail of S_i^* by reflection, and a closed-form expression for the expectation in equation (5) follows,

$$\mathbb{E}[S_i^* | S_i = 0] = S_q - \mathbb{E}[S_i | S_i \geq S_q], \quad (6)$$

where $q = 1 - \Pr(S_i = 0)$ and S_q is the q -th quantile of the empirical distribution of S_i . The right-hand side is fully nonparametric in the observed positive-side distribution. Because $\Pr(S_i = 0) \leq 0.5$ in every subsample in the main analysis, both q and S_q are well defined and the construction does not require any parametric distributional assumption beyond tail symmetry.² If the reflected tail is the wrong shape for the unobserved lower tail, $\mathbb{E}[S_i^* | S_i = 0]$ is mismeasured and the latent discontinuity that scales the correction is off, which Section 5 quantifies by re-estimating under the normal alternative and by varying the implied expectation directly.

Assumption 2 restricts only the *tail* of S_i^* . The mass above S_q on the positive side equals the bunching share $\Pr(S_i = 0)$ by construction, so the reflected lower tail carries mass $\Pr(S_i = 0)$ as well, and the portion of the negative side between $-S_q$ and zero is neither pinned down by the assumption nor used in the construction of $\mathbb{E}[S_i^* | S_i = 0]$ in equation (6). Tail symmetry is also the conservative choice in this application. It implies a more negative $\mathbb{E}[S_i^* | S_i = 0]$ than

²The one exception is the high-GI snack treatment, where 83% of children bunch at zero. There $\Pr(S_i = 0) > 0.5$, the reflected tail is not defined, and the semiparametric-normal specification described below is used instead.

the normal alternative of Section 5, hence a smaller selection correction, and the estimated effect magnitudes are smaller for both outcomes. Figure 6 visualizes the construction by overlaying the observed positive-side density of S_i with its mirror image across zero. Because the mirrored tail is the empirical positive-side tail rather than a fitted normal, it is longer and flatter, so tail symmetry accounts for the extreme latent-demand values to which a fitted normal would assign little mass.

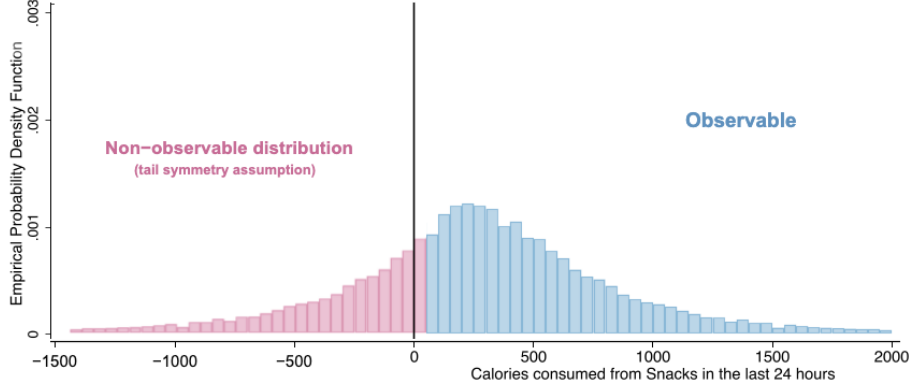


Figure 6: Visualization of the distributional assumption on S_i^* .

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018, author’s calculations.

Notes: The right portion of the figure plots the observed density of S_i for $S_i > 0$. The left portion plots the mirror image of this density across zero, which under Assumption 2 represents the conditional density of S_i^* among children currently bunched at zero. The shaded mean of the mirrored region is the implied value of $\mathbb{E}[S_i^* | S_i = 0]$.

Section 5 also reports estimates under an alternative distributional assumption that substitutes for tail symmetry in the construction of $\mathbb{E}[S_i^* | S_i = 0]$, treating S_i^* as normally distributed and recovering the expectation via a tobit specification.

Under Assumptions 1 and 2, β and δ are identified. Equation (6) delivers $\mathbb{E}[S_i^* | S_i = 0]$ from the observed positive-side distribution, and the control-function regression of equation (5), with the constructed control function $S_i + \mathbb{E}[S_i^* | S_i = 0] \cdot \mathbf{1}(S_i = 0)$ substituting for the unobserved S_i^* , then recovers both slopes. The implementation described next conditions on the observable controls X_i , which relaxes both assumptions by allowing them to hold conditional on X_i rather than unconditionally.

3.4 Estimator and Inference

The estimator implements the two-step identification of Section 3.3 with covariate adjustment. Let X_i denote the vector of pre-treatment controls described in Section 2.3, including age (in years and its square), sex, race, the child’s educational attainment, household poverty income ratio and income tercile, food security, and NHANES cycle fixed effects. Including covariates weakens both identifying assumptions. Assumption 1 needs to hold only for the part of S_i^* orthogonal to X_i , and Assumption 2 needs to hold only conditional on X_i . The conditional versions of the two assumptions

allow the unconditional distribution of S_i^* to be tail-asymmetric and allow the selection slope to vary nonparametrically with X_i . Intuitively, the controls do work that the assumptions would otherwise have to do alone. Comparing a bunched child to a marginal snacker of the same age, household income, food security, and race removes the part of the selection those observables explain, leaving only the residual unobserved variation for tail symmetry and linearity to handle. The wider the set of controls, the smaller that residual.

The estimating equation is

$$Y_i = \beta S_i + m(X_i) + \delta(X_i) \left[S_i + \mathbb{E}[S_i^* | S_i = 0, X_i] \cdot \mathbf{1}(S_i = 0) \right] + \varepsilon_i, \quad (7)$$

where $m(X_i)$ and $\delta(X_i)$ are allowed to depend nonparametrically on X_i and $\mathbb{E}[S_i^* | S_i = 0, X_i]$ is computed within each cell of X_i from the conditional version of equation (6). Because X_i takes too many distinct values to be saturated directly, I discretize X_i into K clusters by k -means clustering on the standardized control vector, and each cluster is then treated as a discrete value of X_i in the saturated specification. The baseline specification uses $K = 50$ for $m(X_i)$ and the conditional expectation, and $K_\delta = 1$ for the selection slope $\delta(X_i)$. Section 5 varies (K, K_δ) over a wide range and shows that the headline estimates are stable.

I implement heterogeneity along an observed dimension, such as the age bin or the household income tercile, by allowing β to vary across the relevant cells. Replacing βS_i in equation (7) with $\sum_g \beta_g S_i \cdot \mathbf{1}(g_i = g)$, where g_i indexes the heterogeneity cell, gives a cell-specific causal effect β_g for each cell. The selection slope $\delta(X_i)$ and the conditional expectation $\mathbb{E}[S_i^* | S_i = 0, X_i]$ continue to depend on the full cluster structure within each cell.

For inference I use a child-level bootstrap that re-runs both the recovery of $\mathbb{E}[S_i^* | S_i = 0, X_i]$ and the estimation of equation (7) on each resample, so the reported standard errors capture variation from both steps. The baseline specification uses 500 bootstrap replications.

4 Results

This section reports the bunching estimates of the causal effect of snack caloric intake on children’s growth and diet quality. Section 4.1 presents the pooled estimates for the two primary outcomes, the BMI z-score and the HEI total score, and decomposes the diet-quality effect across the individual HEI components. Section 4.2 then separates the BMI z-score into height and weight, which is where the body-composition response turns out to live. Section 4.3 evaluates how the two effects vary across age groups, household income terciles, and the composition of the snack, the last covering the sweet-versus-savory split and the high-GI subset. And Section 4.4 explores the complementarity between snacking and the other meals, by estimating the effect of snacking on the amount of calories consumed in non-snack occasions.

Throughout this section, a “snack” is defined as any eating occasion outside of breakfast, lunch, or dinner, as introduced in Section 2.1. Section 5 shows that results are qualitatively similar under the alternative definition that identifies snacks by the food itself rather than by the eating occasion.

4.1 Main Results

Figure 7 reports the bunching estimates of the per-100-kcal effect of snack consumption on the two headline outcomes, the BMI z-score for body composition in the left panel and the HEI total score for diet quality in the right. Each panel shows four estimators, the uncorrected linear regression without and with controls, the semiparametric-normal bunching estimator, and the nonparametric tail-symmetric estimator. The tail-symmetric estimator is the preferred specification because it is the more conservative of the two distributional choices, as discussed in Section 3.3. Point estimates are shown as dots with 95% bootstrapped confidence intervals as horizontal error bars, and the tail-symmetric estimate is highlighted with a larger dot in each panel.

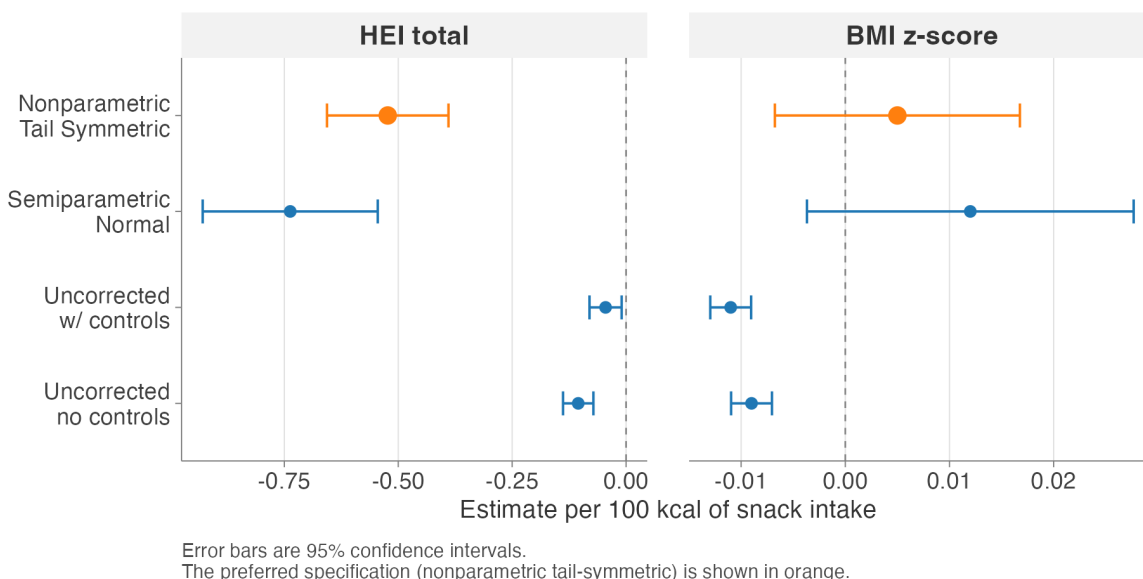


Figure 7: Bunching estimates of the effect of snack caloric intake on the two headline outcomes.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI panel) and 2005–2018 (HEI panel), author’s calculations.

Notes: Each panel shows the four specifications as rows, with the point estimate as a dot and the 95% confidence interval as a horizontal error bar. The preferred nonparametric tail-symmetric specification is shown with a larger dot. Vertical dashed line at zero. Coefficients are per 100 kcal of snack intake.

Confidence intervals are bootstrapped with 500 replications.

There are two main results. First, snacking has a sizable negative effect on diet quality. The nonparametric tail-symmetric specification gives a per-100-kcal HEI loss of 0.52 points ($p < 0.01$), and the semiparametric-normal specification gives a larger loss of 0.74 points ($p < 0.01$). The naive regression coefficient is much smaller in magnitude (-0.05 per 100 kcal). The difference reflects positive selection on unobservables. Children who snack tend to have unobserved characteristics (household food environment, dietary norms) that are independently associated with higher diet quality, so the uncorrected regression understates the true harm by an order of magnitude.

This loss is broad-based across the index rather than concentrated in any single component.

Appendix Figure C.3 reports the per-100-kcal effect on each of the 13 HEI-2015 components, and because the total score is their sum, the component effects add up to the 0.52-point decline. The largest losses fall on whole fruit and total fruit, about 0.12 points each under the tail-symmetric estimator, and on saturated fat, about 0.11, so the marginal snack both crowds out fruit and adds the saturated fat that the moderation components penalize. Added sugar, whole grain, and total vegetables decline more modestly. The two components that move the other way, sodium and refined grain, do so because sweet and fatty snacks dilute those per-energy densities rather than because snacking improves those dimensions of the diet.

Second, snacking has no detectable effect on body composition, and the null is a precise one. The nonparametric tail-symmetric estimate is $\hat{\beta} = 0.005$ z-score units per 100 kcal of snacks (standard error 0.007, 95% confidence interval $[-0.009, 0.019]$), and the semiparametric-normal estimate is 0.012 (standard error 0.010, 95% confidence interval $[-0.007, 0.031]$). The tail-symmetric point estimate is about half a percent of a z-score standard deviation and the semiparametric-normal estimate about one percent. Both confidence intervals are narrow and bracket zero, so this is a tightly identified zero rather than an effect the data are too noisy to detect.

Because the BMI z-score is age-and-sex-standardized, a given z-score change maps to a larger body-weight change for older children, whose CDC reference standard deviations σ_a grow with age (approximately 1.34, 1.98, 2.75, and 2.97 kg/m² at ages 2, 10, 16, and 19). Applying $\Delta\text{weight} = \Delta\text{BMI} \times \text{height}^2$ at a typical height for each age, the point estimate of 0.005 z-score per 100 kcal maps to only about 0.005 kg of body weight per 100 kcal at age 2 (height 0.90 m) and about 0.019 kg at age 10 (height 1.40 m), rising to about 0.040 kg for a typical adolescent at age 16 (height 1.70 m), where the reference standard deviation and body size are largest. Taking that adolescent as the reference case, the upper limit of the 95% confidence interval, 0.019 z-score per 100 kcal, maps to about 0.15 kg per 100 kcal. Applied to the habitual 300-kcal daily snacking level used in the introduction, the point estimate implies about 0.12 kg (0.3 lb) of body weight and even the upper edge of the confidence interval implies only about 0.45 kg (1.0 lb). Because the estimate is a precise zero, these figures bound how large a body-composition response could plausibly be rather than identify one.

4.2 Growth: Height and Weight Separately

The BMI z-score null in Section 4.1 is uninformative about what snacking does to a child’s body, because BMI is a ratio and a ratio can hold fixed while both of its arguments move. Writing $\text{BMI} = W/H^2$, a child who gains mass and stature in the proportion the formula requires registers no change at all. This subsection estimates the effect on height and weight separately and finds exactly that pattern. Snacking makes children taller, and the weight response is what proportional growth would produce rather than what fat accumulation would.

Estimates. Applying the same design to height in centimeters and weight in kilograms gives the results in Table 2. Under the tail-symmetric specification, each 100 kcal of habitual snacking raises

height by 0.148 cm, with a bootstrap standard error of 0.037 and a 95% confidence interval of [0.076, 0.220] that excludes zero comfortably. The weight estimate is 0.108 kg with a standard error of 0.083 and an interval of $[-0.054, 0.270]$ that does not. The semiparametric-normal specification moves both estimates in the same direction, 0.166 cm (0.053) for height and 0.211 kg (0.122) for weight, leaving the height effect significant and the weight effect not. Inference throughout uses 250 child-level bootstrap replications that re-run both estimation steps.

Why BMI does not move. The two estimates reconcile with the BMI z-score null arithmetically rather than coincidentally. Log-differentiating $BMI = W/H^2$ gives $\% \Delta BMI \approx \% \Delta W - 2 \% \Delta H$, so BMI is unchanged whenever weight rises by twice the percentage that height does. Take the age-16 reference child of Section 4.1, at 173.4 cm and 61.6 kg, and the 300 kcal daily snacking level used throughout. Height rises 0.44 cm, or 0.26%, and weight rises 0.32 kg, or 0.53%. Since $0.53 - 2(0.26) = 0.01$, the implied change in BMI is two hundredths of a percent, which is indistinguishable from the 0.005 z-score point estimate the pooled specification returns. The BMI null is therefore not an absence of response. It is the signature of a response that runs along both dimensions in the proportion that leaves the ratio fixed.

What this means. Read as adiposity, the BMI null says snacking does nothing. Read as growth, the same three estimates say something considerably more specific. The marginal snack calorie is being allocated to linear growth and to the mass that accompanies it, not to fat. That is a different claim about children’s bodies and it carries different welfare content, since height in childhood is a standard marker of net nutrition and predicts adult earnings, cognitive test performance, and longevity (Deaton, 2007; Steckel, 2009).

Two readings of the height result are worth separating, because they differ in whether the effect is good news. Under the first, the marginal snack calorie relaxes a binding nutritional constraint and the child grows closer to genetic potential. This is the interpretation the anthropometric literature in economics would apply, and it is more plausible than it first appears even in a well-nourished population, given that the snack piece is dairy-heavy and dairy protein is the most robust dietary predictor of linear growth through the IGF-1 axis (Marshall et al., 2018). Under the second, the extra energy advances growth tempo rather than raising attained stature, accelerating height velocity and bone maturation without changing where the child ends up as an adult. The pediatric endocrinology literature documents this pattern in children with obesity, who are reliably taller through childhood and no taller as adults (Marcovecchio & Chiarelli, 2013). [TO RESOLVE] NHANES is repeated cross-sections and cannot separate the two, since attained adult height is never observed for the same children. The age gradient in the height estimate is the available discriminating evidence, because the tempo story predicts the effect should attenuate and reverse across adolescence while the net-nutrition story does not. Estimating the height effect by age bin is the natural next step.

Composition. [TO COMPUTE] If the height effect runs through dairy protein rather than through energy alone, it should concentrate in the nutrient-dense part of the snack piece rather

than in the sweet or high-glycemic part. The sweet and savory splits and the high-GI subset used in Section ?? apply directly to height and weight and would test this. The prediction is sharp enough to be worth stating in advance. A height effect carried by sweet and high-GI calories favors the insulin and tempo channel, while one carried by dairy and protein-dense calories favors net nutrition.

Table 2: Effect of snack intake on height, weight, and BMI z-score

	Height (cm)	Weight (kg)	BMI z-score
Tail-symmetric	0.148** (0.037)	0.108 (0.083)	0.005 (0.007)
Semiparametric normal	0.166** (0.053)	0.211 (0.122)	0.012 (0.010)

Notes: Estimates are per 100 kcal of habitual snack intake, drinks-inclusive eating-occasion definition. Bootstrap standard errors in parentheses, 250 child-level replications re-running both estimation steps. ** denotes $p < 0.05$.

4.3 Heterogeneity Analysis

Children’s diets are shaped by different decision processes at different ages, and access to alternatives to snack-dense foods varies systematically with household resources. I therefore re-estimate the bunching design separately by age group and by household income tercile. A third cut varies the snack rather than the child. Splitting the treatment by composition contrasts sweet with savory snacks and isolates the high-GI subset, so the question is no longer who is most affected but which kinds of snacks carry the diet-quality damage.

4.3.1 Heterogeneity by Age

I estimate the bunching design separately for the age bins [2, 5], [6, 11], and [12, 19]. The cut-points produce three samples of roughly comparable size while keeping each bin developmentally coherent. Parents largely determine what the youngest children eat at every eating occasion, school-meal and peer environments begin to matter in the middle bin, and adolescents make the bulk of their snack-occasion choices on their own (Savage et al., 2007). I do not choose the cut-points to test a developmental hypothesis. They align with developmental stages, but the design does not identify which features of those stages drive any cross-bin differences. Appendix Figure C.4 plots the estimates for both headline outcomes by age bin across the four specifications of Section 4.1, and the discussion in the body uses the nonparametric tail-symmetric specification throughout.

A caveat about levels comes before the cross-bin comparison. The HEI total score itself falls markedly with age in this sample, from a mean of 54.4 at age 2 to 44.0 at age 19 (a ten-point decline, see Figure 2, left panel). A per-100-kcal HEI penalty of 1 point is therefore a larger share

of the available margin in the older bins, where the baseline HEI is lower, than in the younger bins. Cross-bin comparisons of the absolute per-100-kcal effect should be read with this baseline shift in mind, since part of the apparent attenuation across age bins reflects compression toward a lower HEI baseline rather than a smaller proportional response to snacking.

The HEI estimates fall monotonically across the age bins. The per-100-kcal HEI loss is 1.12 points in the youngest bin (ages 2–5), 0.74 points in the middle bin (ages 6–11), and 0.42 points in the oldest bin (ages 12–19), all significant at conventional levels. One interpretation, consistent with the developmental literature, is that the household food environment of the youngest children concentrates snacks in a particular nutritional profile (low-fruit, high-added-sugar, high-refined-grain), so a marginal increase in S_i^* shifts the daily mix most strongly toward those foods. As children gain autonomy over their snack choices, the harm per kcal attenuates because adolescents substitute toward a wider range of foods that are, on the margin, less concentrated in the lowest-scoring HEI components. The cross-sectional design cannot distinguish this developmental mechanism from the level-baseline mechanism above.

The same caveat motivates a complementary look at the by-age heterogeneity using a diet-quality outcome whose baseline does not move with age the way the raw HEI level does. I re-estimate the design using the *within-age HEI percentile*, defined as each child’s percentile rank of HEI total score computed within their single year of age. By construction the within-age median is 50 at every age, so the age-baseline shift is removed mechanically, and a snack-induced change in this rank measures how far the child moves within the diet-quality distribution of children their own age.

Appendix Figure C.5 shows the raw outcome-discontinuity plot for the within-age HEI percentile, in the same style as Figure 3 of Section 2.2. The jump at the bunching point is visible in the raw data, supporting the design’s identification.

Appendix Table B.3 reports the bunching estimates for the within-age HEI percentile. An additional 100 kcal of snack intake lowers a child’s within-age percentile rank by 1.17 points in the pooled sample, and the gradient remains monotonic across the three age bins (−2.41, −1.64, −0.91 for ages 2–5, 6–11, 12–19). Because this outcome holds the age baseline fixed by construction, the persistence of the gradient shows that the attenuation in the raw-HEI gradient above is a real developmental phenomenon rather than a baseline-compression artifact. The two outcomes in fact attenuate at the same rate: the youngest bin’s effect is 2.65 times the oldest bin’s on the raw HEI score and 2.66 times on the percentile rank, so neutralizing the baseline leaves the gradient intact. Expressed in standard deviations of each outcome, the pooled estimates are likewise indistinguishable, −0.042 for the raw score against −0.041 for the rank. The pattern across income terciles in the same table is roughly flat (−1.26, −1.01, −1.35), so removing the age baseline does not introduce an income gradient.

The BMI z-score point estimates by age bin are small and statistically indistinguishable from zero ($\hat{\beta} = +0.026$, $+0.008$, and -0.006 z-score units per 100 kcal for ages 2–5, 6–11, and 12–19 respectively). Like the pooled estimate in Section 4.1, each is a tightly identified zero rather than a noisy one. Translated to body weight with the age-specific reference standard deviations and

heights of Section 4.1, using ages 2, 10, and 16 as reference points within the three bins, the point estimates correspond to about +0.03 kg (+0.06 lb), +0.03 kg (+0.07 lb), and −0.05 kg (−0.11 lb) of body weight per 100 kcal. The upper limits of the 95% confidence intervals are similarly small, bounding any per-100-kcal effect at about 0.09 kg (0.21 lb) for ages 2–5, 0.13 kg (0.30 lb) for ages 6–11, and 0.06 kg (0.13 lb) for ages 12–19. The body-composition null therefore holds within every age bin, not only in the pooled sample.

4.3.2 Heterogeneity by Household Income

I estimate the bunching design separately by tercile of the household poverty income ratio, the standard NHANES indicator of household economic resources. Higher-income households face a wider set of alternatives to calorie-dense snacks, can absorb the higher relative cost of fresh produce in routine purchasing, and are more likely to organize structured meals around which snacking is residual rather than substitutive. Lower-income households face the opposite set of constraints. The bunching design tests whether these resource differences translate into differential effects of snacking on outcomes. Appendix Figure C.6 plots the estimates for both headline outcomes by tercile across the four specifications of Section 4.1, and the body uses the nonparametric tail-symmetric specification throughout.

The HEI estimates are uniform across income terciles, −0.56 HEI points per 100 kcal in the lowest tercile, −0.48 in the middle, and −0.57 in the highest, all significant at conventional levels. The cross-tercile differences are small relative to their standard errors. Diet-quality damage per kcal of snack consumption therefore appears to be a property of the snack itself, not of the household environment into which it is added. Interventions that reduce children’s snack intake have similar HEI returns regardless of household income.

The BMI z-score point estimates are small and statistically indistinguishable from zero in every income tercile ($\hat{\beta} = 0.012$, -0.004 , and 0.009 z-score units per 100 kcal in the lowest, middle, and highest tercile respectively, with standard errors of 0.009, 0.011, and 0.012). Because the terciles pool all ages, I translate to body weight at the age-16 reference adolescent of Section 4.1, where the point estimates correspond to about +0.10, −0.03, and +0.07 kg per 100 kcal (+0.21, −0.07, and +0.16 lb) and the upper limits of the 95% confidence intervals bound any per-100-kcal effect at about 0.24, 0.14, and 0.26 kg (0.53, 0.32, and 0.57 lb) across the lowest, middle, and highest terciles. Consistent with the pooled null of Section 4.1, no income tercile taken alone shows a detectable effect on the body-composition margin.

4.3.3 Heterogeneity by Snack Composition

The pooled diet-quality effect masks variation across the type of snack a child consumes. I split snack kcal by food type (sweet versus savory) and by glycemic load, re-estimate the design on each piece for both headline outcomes, and compare each to the pooled result. Appendix Figure C.7 plots the estimates for both headline outcomes, together with the other treatment-definition variants examined in Section 5.

The sweet channel accounts for a disproportionate share of the pooled damage. Sweet snacks carry a per-100-kcal HEI loss of 0.85 points against 0.63 for savory, both larger than the pooled 0.52 and significant at conventional levels. The effect is stronger still for high-GI snacks, those with a glycemic index (GI) at or above 70. Because 83% of children consume no high-GI snacks on a given day, the bunching share exceeds the one-half threshold the tail-symmetric estimator requires, and I report the semiparametric-normal specification, under which an additional 100 kcal of high-GI snacking lowers the HEI by 1.49 points, roughly double the pooled semiparametric-normal estimate of 0.74.

The BMI response stays small under every split. The per-100-kcal BMI z-score estimates are +0.019 for sweet, +0.018 for savory, and +0.028 for high-GI snacks. The high-GI interval brackets zero, while the sweet and savory intervals sit marginally above it ($[0.005, 0.033]$ and $[0.002, 0.034]$), a distinction that is statistical rather than substantive. Read at face value, the point estimates translate to about 0.2 kg of adolescent body weight per 100 kcal, the size of the upper bound of the pooled confidence interval, so even where the interval excludes zero the body-composition response is, for any practical purpose, zero. Snack composition reshapes the diet-quality margin, not the body-composition one.

4.4 Snack-Meal Complementarity

Applying the bunching design to non-snack caloric intake, defined as total daily kcal minus snack kcal, tests whether snack calories displace main-meal intake or add to it. Displacement would leave total daily energy roughly unchanged, so snacking would act on diet composition alone. Addition would raise total daily energy, opening an energy-balance channel as well.

The point estimate is positive (0.27 kcal per snack kcal under the nonparametric tail-symmetric specification, $p < 0.01$). Snack calories complement main meals rather than substitute for them, and the familiar story in which a child’s afternoon snack reduces their dinner by a roughly equivalent amount finds no support once selection on household type is netted out. The positive residual is consistent with small physiological habituation, with a behavioral channel the design cannot fully separate from the treatment, or with a confounder that survives the bunching correction. The data do not distinguish among these mechanisms.

Total energy intake therefore rises by $1 + 0.27 = 1.27$ kcal per kcal of snack consumption, a magnitude confirmed by applying the bunching design directly to total daily kcal ($\hat{\beta} = 1.27$, $p < 0.01$). That this additional energy leaves the BMI z-score unmoved is consistent with the pediatric energy-balance literature, in which growth and exercise expenditure absorb a substantial share of marginal caloric intake in middle childhood and adolescence.

The complementarity result speaks to how much children eat at meals when snacking rises. It does not answer whether the composition of the meal piece changes as well. To probe that margin, I apply the bunching design to six per-child intake outcomes measured only on the meal piece of the day, each aggregated over the non-snack rows of the individual food file. Applying HEI-2015 to a sub-day slice is not a standard use of the index, which is calibrated to full-day intake against

per-1000-kcal benchmarks, so I use physical intakes directly. The six outcomes cover both adequacy components (fruit, vegetables, whole grain) and moderation components (refined grain, added sugar, saturated fat). Table B.5 in Appendix B reports the bunching estimate for each, expressed as the percent change in meal-piece intake per 100 kcal of habitual snack intake.

The three adequacy components are statistically flat, with confidence intervals covering zero under either specification, so habitual snacking is not associated with children eating fewer fruits, vegetables, or whole grains at meals. The moderation components are a different story. All three move sharply in the adverse direction, and every effect is significant at the five-percent level. Each additional 100 kcal of habitual snack intake is associated with 1.3% more refined grain, 4.1% more added sugar, and 2.0% more saturated fat at meals under the nonparametric tail-symmetric specification, and with larger effects under the semiparametric-normal alternative. Snacking does not crowd out the healthy dimensions of the meal-time diet, so the adequacy side of the daily HEI does not fall through a spillover mechanism. Instead the meal piece shifts toward the same low-quality moderation profile as the snacks themselves. Combined with the caloric complementarity, the daily HEI penalty from snacking of -0.52 points per 100 kcal is driven by two reinforcing channels rather than by the snack piece alone. The direct channel is that snack foods are themselves lower-density on the moderation components, and the indirect channel is that the meal piece of the day shifts toward the same profile as snacking rises.

The complementarity attenuates monotonically with age, from 0.62 at ages 2–5 to 0.28 at ages 6–11 and 0.24 at ages 12–19, consistent with reduced parental control over portion sizing as children age. The persistence of positive complementarity in the oldest bin indicates that compensatory satiety regulation is incomplete throughout the developmental range studied here. The complementarity is somewhat larger for males (0.27 kcal per snack kcal) than for females (0.20), consistent with the sex-asymmetric partitioning of marginal caloric intake documented in the pediatric body-composition literature.

Splitting snack-occasion kcal by food type into sweet and savory pieces tests whether the meal-time complementarity differs by snack type. Sweet snacks are the more complementary, drawing 0.49 kcal of meal intake per snack kcal (the same value whether sweet is defined by total sugar or by added sugar) against 0.38 for savory snacks, both significant at the one-percent level. The sweet complementarity is largest in adolescence (0.44, 0.36, and 0.57 kcal per kcal at ages 2–5, 6–11, and 12–19), while the savory complementarity is largest among the youngest children (0.62, 0.31, and 0.34). A high-sugar snack triggers a steeper glycemic and insulin response, followed by a rebound hypoglycemia one to two hours later that amplifies meal-time hunger, with insulin sensitivity and appetite signaling more variable in adolescence than earlier in childhood.³

A partition of snack-occasion kcal by Glycemic Index tests this mechanism more directly. High-GI snacks, those with a GI at or above the standard cutoff of 70, trigger the steepest post-snack glucose and insulin response and should therefore generate the largest meal-time complementarity

³Ludwig et al. (1999) find that a high glycemic-index meal lowers late-postprandial glucose and raises subsequent food intake in adolescents, with the mechanism reviewed in Ludwig (2002). Puberty-linked variation in insulin sensitivity is documented by Moran et al. (1999).

(GI values are matched to the NHANES snack food codes using the 2021 International Tables of Glycemic Index and Glycemic Load of Atkinson et al. (2021), with the matching documented in Appendix A.1). The prediction holds. High-GI snacks complement meals at 0.45 kcal per kcal under the semiparametric-normal specification, roughly two-thirds larger than the pooled 0.27, while low-GI snacks complement meals at 0.29, essentially the pooled result. Total energy intake per kcal of high-GI snack is $1 + 0.45 = 1.45$ kcal, against 1.27 for the average snack.

Within ages 12–19, splitting by sex tests whether the sweet-versus-savory contrast tracks puberty-linked metabolic differences. Adolescent females take a larger per-100-kcal HEI loss than males for both snack types, -1.21 against -0.62 for sweet snacks and -0.73 against -0.36 for savory. The energy side runs the other way, with males loading more of the sweet-snack signal into meal calories than females (0.67 against 0.27 kcal per kcal), while the savory complementarity is more even across sex (0.34 against 0.30). The BMI z-score response stays near zero for both sexes and both snack types, between -0.002 and 0.024 per 100 kcal. Halving the sample within each sex limits precision, so the contrast is exploratory evidence on the macronutrient channel rather than identification of a sex-specific causal pathway.

5 Sensitivity and Robustness Checks

This section probes the main estimates of Section 4 along four dimensions, the linearity of the outcome equation in S_i^* near the bunching point (Section 5.1), the parametric distributional assumption on S_i^* (Section 5.2), the role of the cluster-conditional design in relaxing the identifying assumptions (Section 5.3), and the definition of a “snack” itself (Section 5.4). Each subsection takes one of the assumptions stated in Section 3.3 and shows that reasonable relaxations leave the headline estimates qualitatively unchanged.

5.1 Linearity of the Outcome Equation

Equation (3) imposes that the conditional mean of the outcome is linear in S_i^* up to controls, and the identification result reads β off the discontinuity at the bunching point. If linearity fails near zero, the polynomial extrapolation of the positive-side conditional mean back to $S_i = 0^+$ is biased, and so is $\hat{\beta}$.

The standard diagnostic varies the local bandwidth $H_{\max}$ over which the polynomial is fit. When linearity fails locally, the fitted limit at zero depends on how far above zero the positive-side data reach. When the linear approximation is good near the bunching point, the estimate barely moves as $H_{\max}$ contracts toward zero. Figure 8 reports $\hat{\beta}$ as a function of $H_{\max}$ for the BMI z-score (left panel) and the HEI total score (right panel), both under the nonparametric tail-symmetric specification.

At wider bandwidths the point estimates stay close to the headline values in Section 4.1. At the narrowest bandwidths the confidence bands widen substantially and the point estimates oscillate, as expected given how little sample survives once the bandwidth contracts that far. The cluster-

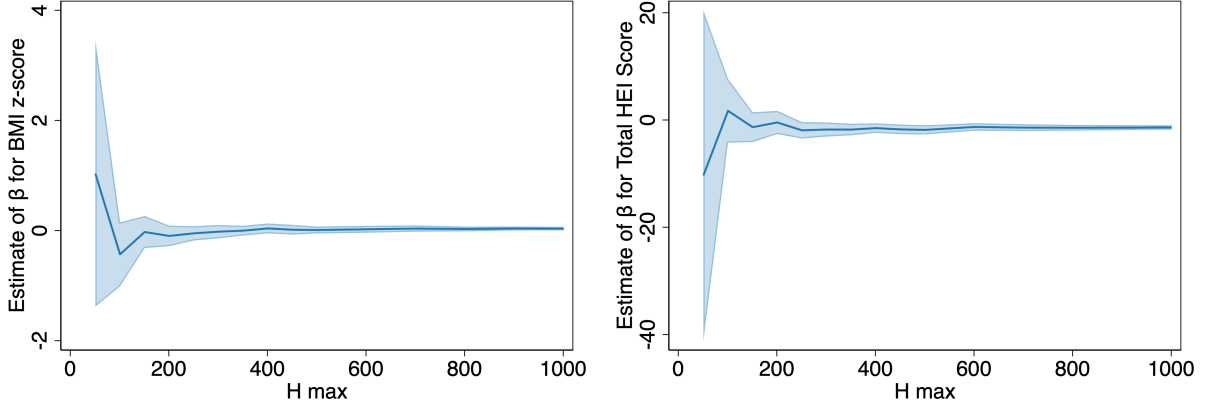


Figure 8: Estimated $\hat{\beta}$ as a function of the local-extrapolation bandwidth $H_{\max}$, under the non-parametric tail-symmetric specification.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI) and 2005–2018 (HEI), author’s calculations.

Notes: The left panel uses the BMI z-score and the right panel uses the HEI total score. Shaded bands are 95% bootstrap confidence intervals from 50 replications.

conditional design of Section 5.3 speaks to the same concern by letting the outcome equation differ across regions of X , and the headline estimates hold up across the range of cluster numbers explored there.

5.2 Distributional Assumption

Section 3.3 maintains the tail-symmetry assumption on S_i^* (Assumption 2), the conservative choice in this application. The lower tail of S_i^* on the negative side mirrors the upper tail of the observed positive distribution above the quantile threshold S_q of equation (6), computed within each covariate cluster, and the middle region between $-S_q$ and zero is left unspecified. Section 4 reports estimates under this assumption alongside the semiparametric-normal alternative, which fits a different shape to the unobserved negative side. For both BMI z-score and HEI, the two specifications agree on sign, and the semiparametric-normal alternative delivers the larger magnitude.

To show that the results do not depend on the exact form of this assumption, Figure 9 reports the estimated $\hat{\beta}$ as the implied $\mathbb{E}[S_i^* \mid S_i = 0, X_i]$ is varied across a range of plausible values, holding the observed outcome discontinuity $\Delta_Y(X_i)$ fixed.

For HEI, moving $\mathbb{E}[S_i^* \mid S_i = 0, X_i]$ across the plotted range shifts the magnitude of $\hat{\beta}$ and preserves sign and significance throughout. For BMI z-score, values close to zero yield a larger confidence interval, while more negative values yield the precise zero estimate reported in Section 4.1.

5.3 Cluster Conditioning and the Identifying Assumptions

The identification in Section 3.3 does not impose linearity (Assumption 1) or tail symmetry (Assumption 2) on the latent snack propensity S_i^* itself. It partitions children into K clusters of the

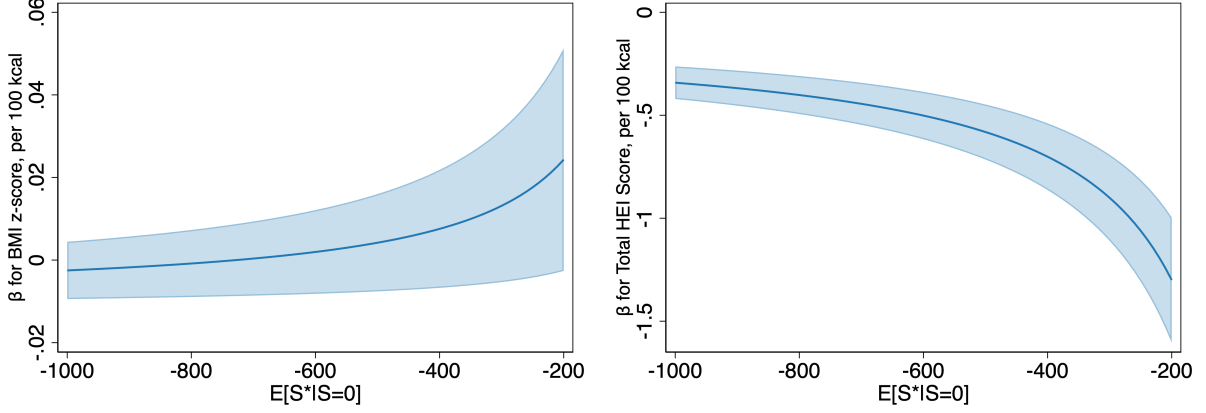


Figure 9: Estimated $\hat{\beta}$ as a function of the assumed value of $\mathbb{E}[S_i^* \mid S_i = 0, X_i]$, holding $\Delta_Y(X_i)$ fixed at its estimated value.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI) and 2005–2018 (HEI), author’s calculations.

Notes: The left panel uses the BMI z-score and the right panel uses the HEI total score. In each panel the two vertical dashed lines mark the implied $\mathbb{E}[S_i^* \mid S_i = 0, X_i]$ under the tail-symmetric specification (*Symmetric*, orange) and the semiparametric-normal specification (*Normal*, grey).

covariate vector X_i and imposes the two assumptions only on the residual variation within a cluster. The conditional mean $\mathbb{E}[S_i^* \mid X_i]$, the outcome intercept $m(X_i)$, the selection slope $\delta(X_i)$, and the dispersion $\sigma(S_i^* \mid X_i)$ are all free to differ from one cluster to the next. Assumption 1 then asks only that the residual $S_i^* - \mathbb{E}[S_i^* \mid X_i]$ enter the outcome linearly, and Assumption 2 only that the standardized residual $\eta_i = (S_i^* - \mathbb{E}[S_i^* \mid X_i]) / \sigma(S_i^* \mid X_i)$ be tail-symmetric. Both are weaker than imposing the same structure on the raw distribution of S_i^* , because the cluster effects absorb the differences across children in the level and spread of snacking that would otherwise have to satisfy the assumptions on their own.

For this conditioning to do any work, the clusters must differ in their snacking distributions in the first place. Figure 10 shows that they do. The within-cluster median of S_i rises steadily across the fifty clusters, from roughly 250 to 500 kcal per day, and the interquartile ranges plotted for selected clusters, which run from under 200 to more than 600 kcal, show that the spread of S_i varies across clusters as much as its level does. This is the heterogeneity that the cluster effects absorb. A single-cluster implementation would instead leave it in the residual, where it would strain both the linearity and the tail-symmetry requirements.

The second diagnostic asks whether the amount of conditioning changes the answer. Figure 11 varies the number of clusters from one, the unconditional case that leaves all heterogeneity in the residual, up to $K = 60$, re-estimating $\hat{\beta}$ at each value under the nonparametric tail-symmetric specification. The estimate barely moves across the whole range and is flat beyond the first several clusters. If the heterogeneity the clusters absorb were biasing the estimate, $\hat{\beta}$ would keep shifting as the conditioning grew finer. Its stability means that imposing the assumptions on the within-cluster residual, rather than on the raw S_i^* , recovers the same headline effect reported in Section 4.1, and

that the effect does not turn on how finely the design conditions on X_i .

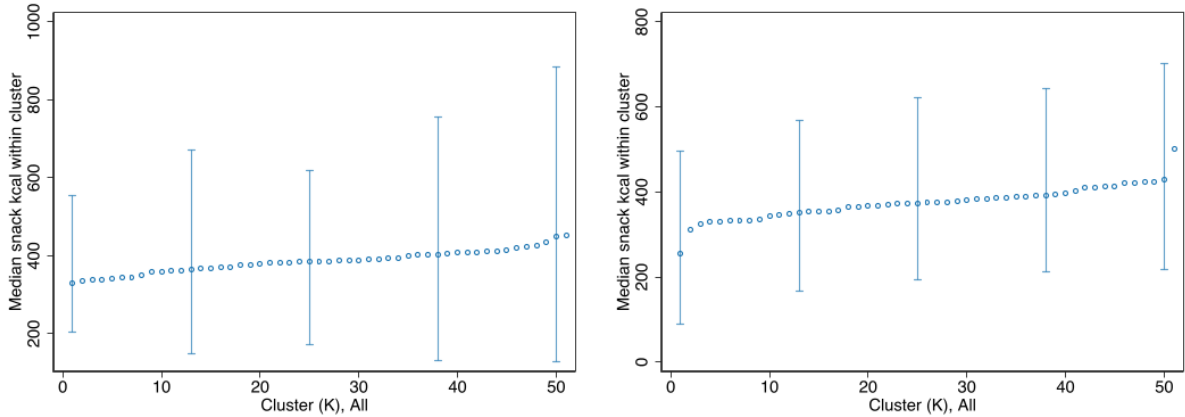


Figure 10: Heterogeneity in the distribution of S_i across clusters.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI) and 2005–2018 (HEI), author’s calculations.

Notes: Each panel shows the median of S_i within each of the $K = 50$ clusters of X , sorted from left to right by the median. The interquartile ranges within selected clusters are also shown. The left panel uses the BMI sample and the right panel uses the HEI sample.

5.4 Alternative Snack Definition

The main definition in Section 4 treats snacking as an eating-occasion event. Any food or calorie-bearing beverage consumed outside the three main meals counts as snack kcal, following the nutritional convention (Hess et al., 2016). A conceptually distinct alternative identifies snacks by the food itself rather than by the occasion. Under this food-classification view, cookies, chips, candy, sweetened cereal, ice cream, and other items whose USDA simplified category is “Snacks” contribute to snack kcal whenever they are eaten, so a bag of chips at lunchtime counts and an apple at 3pm does not. The two definitions agree on most observations and disagree at the margin. The eating-occasion definition catches calorie-bearing off-meal drinks and calorie-dense meal-time foods eaten at non-meal times (a sandwich at 3pm), which the food-classification definition misses. The food-classification definition catches packaged snack foods eaten alongside meals, which the eating-occasion definition misses.

Under the food-classification definition, 12.6% of children in the BMI sample consume zero kcal from snack-classified foods on the previous day, close to the 11.1% bunching share under the eating-occasion definition, and among the children who do consume them, snack-classified foods average 421 kcal, against the 563 eaten at snack occasions under the eating-occasion definition. The food-classification bunching point is therefore informative in the same sense as the eating-occasion one. Appendix Figure C.8 reproduces the raw outcome-discontinuity plot of Figure 3 for the food-classification sample. The same qualitative jumps appear at the bunching point under this alternative definition.

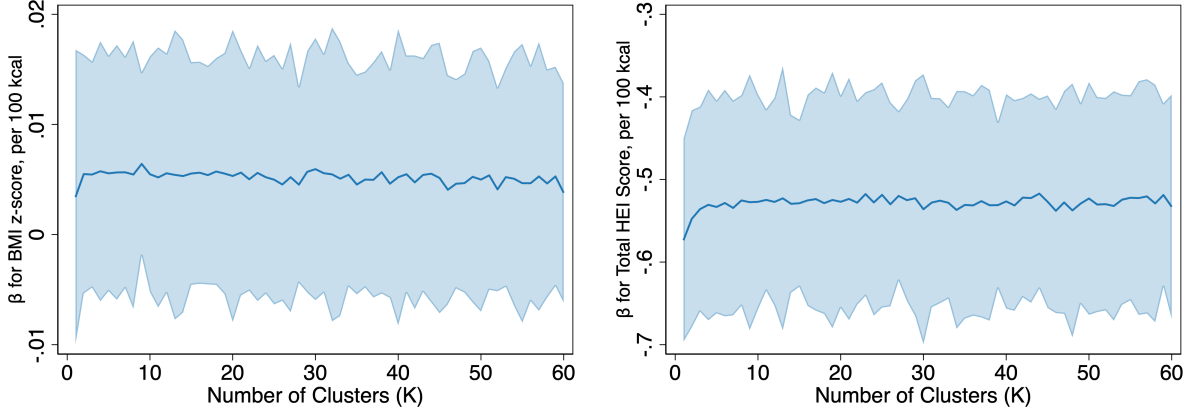


Figure 11: Estimated $\hat{\beta}$ as a function of the number of clusters K used in the residualization, under the nonparametric tail-symmetric specification.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI) and 2005–2018 (HEI), author’s calculations.

Notes: The left panel uses the BMI z-score and the right panel uses the HEI total score. Shaded bands are 95% bootstrap confidence intervals from 50 replications.

Table 3 re-estimates the main and heterogeneity specifications under the food-classification definition. The HEI effect stays negative and significant, with similar magnitudes across the two definitions. Under the semiparametric-normal specification the per-100-kcal HEI loss is -0.878 under food classification and -0.737 under the eating occasion. The BMI z-score point estimate is small and statistically indistinguishable from zero under both (0.010 food classification, 0.012 eating occasion). The age gradient on HEI attenuates monotonically under both definitions, and the youngest-bin estimate is even more negative under food classification (-2.37 vs. -1.80), which fits the food-classification definition placing more snack-kcal weight on the high-added-sugar packaged foods that appear disproportionately in younger children’s intake.

One heterogeneity pattern does shift between the two definitions. HEI estimates by income tercile are roughly uniform under the main definition (-0.76 , -0.69 , -0.81) but fall monotonically under the food-classification definition (-0.53 , -0.93 , -1.20). The divergence likely comes from the mechanics of the food-classification definition, which excludes high-caloric meal-time foods eaten at snack occasions. In higher-income households, snack-classified foods may be unusually concentrated in low-HEI-scoring components (ultra-processed items served as side dishes at meals, for instance), which inflates the apparent HEI gradient under the food-classification definition without any corresponding shift under the eating-occasion definition. I read the difference as a feature of food categorization rather than a change in the underlying behavior, and I prefer the eating-occasion definition on conceptual grounds because it follows the nutritional-literature convention.

Panels D and E extend the raw-BMI heterogeneity to age bins and income terciles. The point estimates are small and statistically indistinguishable from zero in every cell under both definitions, in line with the pooled finding of a precise zero body-composition response at the population scale.

Table 3: Comparison of bunching estimates ($\hat{\beta}$, semiparametric-normal specification) under the main eating-occasion definition (drinks-inclusive) and the alternative food-classification definition of a snack.

Outcome / Subsample	Main (eating occasion)	Robustness (food type)
<i>Panel A: pooled</i>		
BMI z-score	0.012 (0.010)	0.010 (0.011)
HEI total	−0.737** (0.098)	−0.878** (0.131)
<i>Panel B: HEI by income tercile</i>		
Lowest tercile	−0.760** (0.165)	−0.533** (0.205)
Middle tercile	−0.688** (0.182)	−0.927** (0.251)
Highest tercile	−0.808** (0.192)	−1.200** (0.235)
<i>Panel C: HEI by age bin</i>		
Ages 2–5	−1.795** (0.524)	−2.370** (0.559)
Ages 6–11	−1.128** (0.231)	−0.824** (0.248)
Ages 12–19	−0.575** (0.107)	−0.612** (0.152)
<i>Panel D: BMI (kg/m²) by age bin</i>		
Ages 2–5	−0.019 (0.074)	−0.050 (0.080)
Ages 6–11	−0.012 (0.085)	−0.153 (0.093)
Ages 12–19	−0.002 (0.046)	0.076 (0.064)
<i>Panel E: BMI (kg/m²) by income tercile</i>		
Lowest tercile	0.086 (0.084)	−0.026 (0.089)
Middle tercile	−0.094 (0.065)	0.124 (0.098)
Highest tercile	0.130 (0.087)	0.090 (0.104)

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI panel) and 2005–2018 (HEI panel), author’s calculations.

Notes: BMI z-score, raw BMI (kg/m²), and HEI total-score coefficients are reported per 100 kcal. Age-bin and income-tercile heterogeneity is reported in raw BMI to preserve the cross-bin comparability that the age-and-sex standardization would obscure within narrow subsamples. Standard errors in parentheses are bootstrap standard errors.

** $p < 0.05$, * $p < 0.10$.

Appendix Figure C.7 collects the treatment-definition variants discussed throughout the paper (the sweet and savory splits, the high-GI subset, the food-classification definition, and the mean-imputation variant for missing controls) into a single specification curve alongside the main estimate. The diet-quality effect is negative and significant in every row, and every BMI estimate stays within 0.03 z-score units of zero.

6 Conclusion

This paper estimates the causal effect of snacking on body composition and diet quality among American children aged 2 to 19. The empirical strategy exploits the bunching of children at zero snack consumption to identify the causal impact of latent snack demand on the outcomes. Because consumption cannot fall below zero, this natural lower bound generates a mass point at the origin that concentrates unobserved confounders. Following Caetano et al. (2024b), Caetano et al. (forthcoming) this approach applies the corner-bunching framework to a behavioral setting in which the bunching point is generated by a mechanical constraint on the data rather than by an institutional kink in a choice set.

Snack consumption substantially lowers diet quality without altering child body composition. Under the nonparametric tail symmetric specification, every 100 kcal of snack intake lowers the HEI total score by 0.52 points, a statistically significant drop that remains consistent across age groups and household income terciles. In contrast, the estimated effect on BMI z-scores is 0.005 units per 100 kcal with a standard error of 0.007 (95% confidence interval $[-0.009, 0.019]$). Because the upper bound of this interval corresponds to an increase of just 0.15 kg (0.33 lb) in adolescent body weight per 100 kcal, the results establish a precisely identified null effect on weight rather than an imprecise estimate masked by noise.

On the diet-quality side, the loss attenuates monotonically with age, from -1.12 points per 100 kcal in ages 2–5 to -0.42 points per 100 kcal in ages 12–19, consistent with younger children’s food environments being more concentrated on a narrow set of low-HEI snack foods that carry a larger marginal harm. The BMI point estimates by age bin and by income tercile are small and statistically indistinguishable from zero throughout, so the null pooled body-composition result is not masking a subgroup effect. Among adolescents, the cross-sex contrast for sweet snacks shows that females take a larger HEI loss than males, an exploratory pattern that invites further work on the partitioning of marginal calories during puberty.

Because the pooled coefficient estimates the price of a marginal calorie, scaling this effect generates a counterfactual scenario where snack consumption is eliminated entirely. On that reading, removing 300 kcal of daily snacking produces about 1.6 HEI points. However, the per calorie penalty is not constant across snacks. Sweet snack cost 0.85 points per 100 kcal against 0.63 for savory snacks. Similarly, under the semiparametric normal specification, high glycemic calories cost 1.49 points per 100 kcal, compared to 0.74 points for average snack calories under that same specification. What children eat therefore matters significantly more for diet quality than total caloric intake, indicating that guiding snack substitution holds far greater practical potential than targeting complete abstention.

To test the potential impact of guided substitution, I construct an isocaloric counterfactual for the 19,073 snacking children in the HEI sample. For this exercise, I reallocate calories by removing sweet and salty items, including cookies, cakes, salty grain snacks, candy, ice cream, sugar sweetened beverages, sports drinks, and potato chips, and I replace these items with an equal number of calories drawn from a nutrient dense basket of foods that children already eat as snacks,

weighted toward whole fruit, plain yogurt, natural cheese, unsalted nuts, air popped popcorn, and whole grain crackers. This swap reallocates 319 kcal per child, representing 58 percent of mean snack calories, while holding total daily energy intake fixed. Rescoring overall diet quality raises average HEI scores from 47.7 to 57.3, a gain of 9.7 points that increases with age from 7.6 points among children aged 2–5 to 10.9 points among adolescents aged 12–19. Table B.4 reports the exercise by age bin together with its decomposition across the HEI components. Added sugars and whole fruit each supply a little over two points of the total, with total fruit contributing 1.8, dairy 1.6, and refined grains 1.1.

This calculation represents an accounting exercise rather than a causal estimate. Because it assumes perfect compliance, ignores behavioral adjustments, and relies on a baseline food basket rather than empirically estimated substitution patterns, the 9.7 point gain is best interpreted as an upper bound on what compositional change can deliver. Set against the 1.6 points available from removing 300 kcal, this benchmark highlights where the actual policy leverage lies. The exercise also aligns naturally with the null finding for body composition, as holding total daily energy intake constant should not alter child weight.

The estimates sit alongside a set of established policies that target children’s diet quality through different margins, as summarized in Table B.6. The National School Lunch and School Breakfast Programs reach approximately 30 million children daily at a combined federal cost above \$20 billion a year, but by design they modify only structured meal occasions rather than snacking behavior (Hinrichs, 2010; Schanzenbach, 2009). Excise taxes on sugar-sweetened beverages reduce taxed-beverage purchases at minimal fiscal cost but do not act on the decision to snack at all (Allcott et al., 2019; Cawley & Frisvold, 2017). Meanwhile, SNAP represents the largest federal program at about \$120 billion annually, but it subsidizes food purchases without restricting their composition or their timing across eating occasions (H. W. Hoynes & Schanzenbach, 2009; Kreider et al., 2012). Consequently, standard policy instruments leave unstructured eating outside meals largely unaddressed. The bunching estimates quantifies that margin, finding that snack intake reduces total daily HEI by 0.52 points per 100 kcal overall and by 1.12 points for ages 2–5. For scale, the post-HHFKA school-meal reforms raised the whole-day HEI of participants by 1.9 to 4.7 points (Kinderknecht et al., 2020). While a full cost-benefit comparison requires matched effect estimates, the margin least reached by the existing programs is also the largest per-kcal driver of daily diet quality identified here, emphasizing its importance in structuring public nutrition policy.

These findings point to several extensions for future research. Longitudinal panel data tracking individual dietary trajectories over childhood could identify the dynamic process through which habitual snacking influences body composition, though pursuing this direction requires new survey efforts capable of following child diets over time. Linking the bunching estimate to policy-relevant variation in the snacking environment, including school meal reforms, SNAP eligibility expansions, and excise taxes on sugar-sweetened products, would test whether the implied causal effects survive in institutionally induced variation and would provide direct guidance for intervention design. The bunching design’s contribution to that agenda is to fix the population-scale magnitudes of the snack-

to-outcome relationships, establishing the prior against which any policy-induced reduction in snack intake can be evaluated.

References

- Allcott, H., Lockwood, B. B., & Taubinsky, D. (2019). Should we tax sugar-sweetened beverages? an overview of theory and evidence. *Journal of Economic Perspectives*, 33(3), 202–227.
- Almond, D., & Currie, J. (2011). Human capital development before age five. In *Handbook of labor economics* (pp. 1315–1486).
- Anderson, M. L., Gallagher, J., & Ramirez Ritchie, E. (2018). School meal quality and academic performance. *Journal of Public Economics*, 168, 81–93. <https://doi.org/10.1016/j.jpubeco.2018.09.013>
- Atkinson, F. S., Brand-Miller, J. C., Foster-Powell, K., Buyken, A. E., & Goletzke, J. (2021). International tables of glycemic index and glycemic load values 2021: A systematic review. *The American Journal of Clinical Nutrition*, 114(5), 1625–1632. <https://doi.org/10.1093/ajcn/nqab233>
- Birch, L. L. (1999). Development of food preferences. *Annual Review of Nutrition*, 19(1), 41–62.
- Bleakley, H. (2007). Disease and development: Evidence from hookworm eradication in the American South. *Quarterly Journal of Economics*, 122(1), 73–117.
- Bütikofer, A., Mølland, E., & Salvanes, K. G. (2018). Childhood nutrition and labor market outcomes: Evidence from a school breakfast program. *Journal of Public Economics*, 168, 62–80. <https://doi.org/10.1016/j.jpubeco.2018.08.008>
- Caetano, C., Caetano, G., Mazetto, D., & Skira, M. (2025). *The effect of video watching on children's cognitive skills* [Working paper.].
- Caetano, C., Caetano, G., & Nielsen, E. R. (2024a). *Are children spending too much time on enrichment activities?* [Working paper.].
- Caetano, C., Caetano, G., & Nielsen, E. R. (2024b). Correcting for endogeneity in models with bunching. *Journal of Business & Economic Statistics*, 42(3), 851–863.
- Caetano, C., Caetano, G., Nielsen, E. R., & Sanfelice, V. (forthcoming). *The effect of maternal labor supply on children*. [Forthcoming, Journal of Labor Economics].
- Caetano, C., Caetano, G., & Santos, K. (2026). *Household chores and the gender gap in child development: Evidence from bunching* [Working paper.].
- Cawley, J. (2010). The economics of childhood obesity. *Health Affairs*, 29(3), 364–371.
- Cawley, J., & Frisvold, D. E. (2017). The pass-through of taxes on sugar-sweetened beverages to retail prices: The case of Berkeley, California. *Journal of Policy Analysis and Management*, 36(2), 303–326.
- Cawley, J., & Meyerhoefer, C. (2012). The medical care costs of obesity: An instrumental variables approach. *Journal of Health Economics*, 31(1), 219–230.
- Cunha, F., & Heckman, J. J. (2007). The technology of skill formation. *American Economic Review*, 97(2), 31–47.
- Currie, J., DellaVigna, S., Moretti, E., & Pathania, V. (2010). The effect of fast food restaurants on obesity and weight gain. *American Economic Journal: Economic Policy*, 2(3), 32–63.
- Deaton, A. (2007). Height, health, and development. *Proceedings of the National Academy of Sciences*, 104(33), 13232–13237.
- Della Corte, K. A., Bosler, T., McClure, C., Buyken, A. E., Della Corte, D., & Titcomb, T. J. (2024). Development of a national database for dietary glycemic index and load for nutritional epidemiologic studies in the United States. *The American Journal of Clinical Nutrition*.
- Handbury, J., Rahkovsky, I., & Schnell, M. (2015). *Is the focus on food deserts fruitless? Retail access and food purchases across the socioeconomic spectrum* [Working Paper.].
- Heckman, J. J. (2007). The economics, technology, and neuroscience of human capability formation. *Proceedings of the National Academy of Sciences*, 104(33), 13250–13255.

- Hess, J. M., Jonnalagadda, S. S., & Slavin, J. L. (2016). What is a snack, why do we snack, and how can we choose better snacks? a review of the definitions of snacking, motivations to snack, contributions to dietary intake, and recommendations for improvement. *Advances in Nutrition*, 7(3), 466–475.
- Hinrichs, P. (2010). The effects of the national school lunch program on education and health. *Journal of Policy Analysis and Management*, 29(3), 479–505.
- Hoynes, H., Schanzenbach, D. W., & Almond, D. (2016). Long-run impacts of childhood access to the safety net. *American Economic Review*, 106(4), 903–934.
- Hoynes, H. W., & Schanzenbach, D. W. (2009). Consumption responses to in-kind transfers: Evidence from the introduction of the Food Stamp Program. *American Economic Journal: Applied Economics*, 1(4), 109–139.
- Kinderknecht, K., Harris, C., & Jones-Smith, J. (2020). Association of the Healthy, Hunger-Free Kids Act with dietary quality among children in the US National School Lunch Program. *JAMA*, 324(4), 359–368.
- Kleven, H. J. (2016). Bunching. *Annual Review of Economics*, 8, 435–464.
- Kreider, B., Pepper, J. V., Gundersen, C., & Jolliffe, D. (2012). Identifying the effects of SNAP on child health outcomes when participation is endogenous and misreported. *Journal of the American Statistical Association*, 107(499), 958–975.
- Ludwig, D. S. (2002). The glycemic index: Physiological mechanisms relating to obesity, diabetes, and cardiovascular disease. *JAMA*, 287(18), 2414–2423.
- Ludwig, D. S., Majzoub, J. A., Al-Zahrani, A., Dallal, G. E., Blanco, I., & Roberts, S. B. (1999). High glycemic index foods, overeating, and obesity. *Pediatrics*, 103(3), e26.
- Marcovecchio, M. L., & Chiarelli, F. (2013). Obesity and growth during childhood and puberty. *World Review of Nutrition and Dietetics*, 106, 135–141.
- Marshall, T. A., et al. (2018). Higher longitudinal milk intakes are associated with increased height in a birth cohort followed for 17 years. *The Journal of Nutrition*.
- Micha, R., Peñalvo, J. L., Cudhea, F., Imamura, F., Rehm, C. D., & Mozaffarian, D. (2017). Association between dietary factors and mortality from heart disease, stroke, and type 2 diabetes in the United States. *JAMA*, 317(9), 912–924.
- Mikkilä, V., Räsänen, L., Raitakari, O. T., Pietinen, P., & Viikari, J. (2005). Consistent dietary patterns identified from childhood to adulthood: The Cardiovascular Risk in Young Finns Study. *British Journal of Nutrition*, 93(6), 923–931.
- Moran, A., Jacobs, D. R., Steinberger, J., Hong, C.-P., Prineas, R., Luepker, R., & Sinaiko, A. R. (1999). Insulin resistance during puberty: Results from clamp studies in 357 children. *Diabetes*, 48(10), 2039–2044.
- National Cancer Institute, Division of Cancer Control and Population Sciences. (2024). Healthy eating index (hei) components and scoring standards [Accessed 2026-05-22].
- Piernas, C., & Popkin, B. M. (2010). Trends in snacking among U.S. children. *Health Affairs*, 29(3), 398–404.
- Saez, E. (2010). Do taxpayers bunch at kink points? *American Economic Journal: Economic Policy*, 2(3), 180–212.
- Savage, J. S., Fisher, J. O., & Birch, L. L. (2007). Parental influence on eating behavior: Conception to adolescence. *Journal of Law, Medicine & Ethics*, 35(1), 22–34.
- Schanzenbach, D. W. (2009). Do school lunches contribute to childhood obesity? *Journal of Human Resources*, 44(3), 684–709.
- Schwingshackl, L., Bogensberger, B., & Hoffmann, G. (2018). Diet quality as assessed by the Healthy Eating Index, alternate Healthy Eating Index, Dietary Approaches to Stop Hypertension

- score, and health outcomes: An updated systematic review and meta-analysis of cohort studies. *Journal of the Academy of Nutrition and Dietetics*, 118(1), 74–100.
- Steckel, R. H. (2009). Heights and human welfare: Recent developments and new directions. *Explorations in Economic History*, 46(1), 1–23.
- Steinberg, L., Graham, S., O'Brien, L., Woolard, J., Cauffman, E., & Banich, M. (2009). Age differences in future orientation and delay discounting. *Child Development*, 80(1), 28–44.
<https://doi.org/10.1111/j.1467-8624.2008.01244.x>

A Construction of the High-GI Snack Treatment

A.1 Construction of the High-GI Snack Treatment

This subsection documents the construction of the high-GI snack treatment shown in the specification curve (Figure C.7). The treatment is the total daily kcal consumed at snack occasions from foods whose Glycemic Index (GI) equals or exceeds 70, the standard high-GI cutoff in the nutrition literature. The construction requires (i) a per-food GI database whose methodology is defensible, and (ii) a mapping between that database and NHANES eight-digit food codes.

Source of GI values. I use the 2021 International Tables of Glycemic Index and Glycemic Load compiled by Atkinson et al. (2021), the standard nutrition-epidemiology reference for GI values. Supplemental Table 1 of that publication contains 2,091 per-food entries with GI and Glycemic Load (GL) values tested under the ISO 26642:2010 methodology, which restricts studies to ≥ 10 healthy adults aged 18–65 with normal glucose tolerance and standardized sampling timepoints. Supplemental Table 2 adds 1,927 entries collected under less strict methodology. I use both tables, prioritizing Supplemental Table 1 when a match exists in both, for a combined pool of roughly 4,000 per-food GI values.

Mapping to NHANES food codes. The Atkinson tables are keyed on per-product descriptions with brand names and countries of origin (e.g., “Apple Blueberry muffin (Sara Lee Bakery, Australia)”), not on the eight-digit NHANES food codes. Producing the mapping is therefore the main data-engineering task of the exercise. I follow the general approach of Della Corte et al. (2024) but adapt the pipeline to work without their proprietary crosswalk (which is not publicly released and requires an executed data-use agreement).

Concretely, I first pull the per-cycle food-code-to-description tables from NHANES (DRXFCD_C through DRXFCD_J) and union them across the 2003–2018 cycles, retaining the most recent description when a food code appears in multiple cycles. This produces a 10,977-entry NHANES food-code lookup. I then restrict to the 3,356 food codes that actually appear at snack occasions in the analytic sample under the drinks-inclusive snack definition.

For each snack food code, I score every Atkinson entry in the aligned Atkinson category by a composite of $2 \times$ (first-token match) plus token overlap between the two descriptions. A first-token match is present when the leading content word of the NHANES description (e.g., “apple” in “Apple, raw”) appears anywhere in the Atkinson item text. When the top score exceeds 1.5, I assign the GI of the best Atkinson entry (labeled “direct match”). When the top score is below 1.5 or no candidates exist, I assign the mean GI of the primary aligned Atkinson category (labeled “category-mean fallback”). Food codes in NHANES categories that have no natural Atkinson alignment (Animal Protein, Fats and Oils, Condiments, and a small residual of missing categories) are assigned $GI = 0$; this is defensible because these foods carry essentially no available carbohydrate and therefore produce no glycemic response.

The pipeline directly matches 1,693 of the 3,356 snack food codes (50.4%), covering 70.0% of the analytic-sample snack kcal. Category-mean fallback covers another 1,144 codes (25.6% of snack kcal), and 519 codes (4.4% of kcal) are left unmatched at $GI = 0$. The dominant sources of imperfect direct matches on the top-mass codes are truncated NHANES descriptions (“Milk, whole”) that resolve to related-but-not-identical Atkinson entries (“Milk, condensed, sweetened”) rather than to the ideal product-specific reference. Because the treatment aggregates to a daily total across many per-food rows and then classifies at the $GI \geq 70$ threshold, per-code point-estimate error at intermediate GI values leaves the treatment unchanged unless the error straddles the threshold. The treatment is therefore robust to the kind of per-food GI noise the pipeline produces.

Treatment construction. For each per-food row in the individual food file, I compute the Glycemic Load as

$$GL_{\text{food}} = GI \times \frac{\max(0, \text{carb}_g - \text{fiber}_g)}{100},$$

where $\text{carb}_g - \text{fiber}_g$ is the standard nutrition-epidemiology approximation to available carbohydrate. Aggregating to per-child daily totals restricted to snack rows produces the two candidate treatments used in the paper. The continuous variant is

$$\text{snack_gl_daily} = \sum_{\text{snack rows}} GL_{\text{food}},$$

with sample mean 39.6 and a 2.5% bunching share at zero. The binary-classification variant, which is the treatment used in the specification curve, is

$$\text{high_gi_snack_kcal} = \sum_{\text{snack rows, } GI \geq 70} \text{kcal}_{\text{food}},$$

with sample mean 32.7 kcal and an 82.9% bunching share at zero. The 82.9% mass at zero exceeds the tail-symmetric threshold of 50%, so I report the HetTobit semiparametric-normal estimator as the main coefficient and the HetSymm nonparametric tail-symmetric estimator as reference, following the convention used elsewhere in the paper when the bunching share at zero exceeds one-half.

B Tables

Table B.1: HEI-2020 components and scoring standards (ages 2 and older).

Component	Max. points	Standard for maximum score	Standard for minimum score of zero
<i>Adequacy components (higher intake → higher score)</i>			
Total Fruits ^a	5	≥ 0.8 cup equiv. / 1,000 kcal	No fruits
Whole Fruits ^b	5	≥ 0.4 cup equiv. / 1,000 kcal	No whole fruits
Total Vegetables ^c	5	≥ 1.1 cup equiv. / 1,000 kcal	No vegetables
Greens and Beans ^c	5	≥ 0.2 cup equiv. / 1,000 kcal	No dark-green vegetables or legumes
Whole Grains	10	≥ 1.5 ounce equiv. / 1,000 kcal	No whole grains
Dairy ^d	10	≥ 1.3 cup equiv. / 1,000 kcal	No dairy
Total Protein Foods ^c	5	≥ 2.5 ounce equiv. / 1,000 kcal	No protein foods
Seafood & Plant Proteins ^{c,e}	5	≥ 0.8 ounce equiv. / 1,000 kcal	No seafood or plant proteins
Fatty Acids ^f	10	(PUFAs + MUFAs) / SFAs ≥ 2.5	(PUFAs + MUFAs) / SFAs ≤ 1.2
<i>Moderation components (higher intake → lower score, reverse-coded)</i>			
Refined Grains	10	≤ 1.8 ounce equiv. / 1,000 kcal	≥ 4.3 ounce equiv. / 1,000 kcal
Sodium	10	≤ 1.1 gram / 1,000 kcal	≥ 2.0 grams / 1,000 kcal
Added Sugars ^g	10	≤ 6.5% of energy	≥ 26% of energy
Saturated Fats	10	≤ 8% of energy	≥ 16% of energy

Intakes between the minimum and maximum standards are scored proportionally. ^a Includes 100% fruit juice. ^b Includes all forms except juice. ^c Includes legumes (beans and peas). Under HEI-2020, legumes are counted in both the vegetable group (Total Vegetables and Greens and Beans) and, when total protein intake from animal-source foods is below the recommendation, in the protein group (Total Protein Foods and Seafood and Plant Proteins). ^d Includes all milk products, such as fluid milk, yogurt, and cheese, and fortified soy beverages. ^e Includes seafood, nuts, seeds, and soy products (other than beverages). ^f Ratio of poly- and monounsaturated fatty acids (PUFAs and MUFAs) to saturated fatty acids (SFAs). ^g Includes caloric sweeteners and syrups used as sweeteners in other food products, as well as sugars added in food preparation, processing, and added at the table.

Source: National Cancer Institute, Division of Cancer Control and Population Sciences (2024).

Notes: The upper block lists the “adequacy” components for which a higher intake earns a higher score. The lower block lists the “moderation” components, which are reverse-coded so that a lower intake earns a higher score. Intakes between the minimum and maximum standards are scored proportionally. The cut-offs match HEI-2015. The substantive update introduced with HEI-2020 is the reclassification of beans and peas as legumes that are counted in both the vegetable and the protein components.

Table B.2: Snack categories consumed at snack occasions, ranked by prevalence among snacking children, with the average kcal consumed among the children who consume each category.

Rank	Simplified category	% of snackers	Avg kcal
1	Snacks	79.4%	325
2	Beverages	69.0%	111
3	Fruits and Vegetables	38.9%	128
4	Dairy and Eggs	32.1%	180
5	Grains	17.0%	174
6	Mixed Dishes	9.7%	403
7	Animal Protein	8.7%	177
8	Condiments and Sauces	6.9%	33
9	Fats and Oils	5.9%	88
10	Sugar	5.7%	68
11	Plant-based Protein	5.5%	206
12	Baby Food	0.4%	192

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018, individual food file (day-1 recall), author’s calculations.

Notes: Snack occasions use the drinks-inclusive eating-occasion definition, under which off-meal calorie-bearing drinks count as snacks. “% of snackers” is the share of the 22,965 children with any snack who consume the category at a snack occasion. “Avg kcal” is the mean day-1 kcal from the category among those consumers, not a total over the sample. Referenced from Section 2.2.

Table B.3: Bunching estimates of the effect of snack caloric intake on the within-age HEI percentile.

Subsample	Within-age HEI pctl
<i>Panel A: pooled</i>	
All (N=19,964)	−1.17** (0.17)
<i>Panel B: by age bin</i>	
Ages 2–5 (N=4,776)	−2.41** (0.78)
Ages 6–11 (N=6,839)	−1.64** (0.44)
Ages 12–19 (N=8,349)	−0.91** (0.20)
<i>Panel C: by income tercile</i>	
Lowest (N=6,679)	−1.26** (0.28)
Middle (N=6,780)	−1.01** (0.26)
Highest (N=6,505)	−1.35** (0.35)

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, HEI sample, author’s calculations.

Notes: The within-age HEI percentile is the rank of the HEI total score within each single year of age (median = 50 by construction at every age). Coefficients are per 100 kcal of snack consumption. Standard errors in parentheses are bootstrap standard errors based on 100 replications. Nonparametric tail-symmetric specification throughout, drinks-inclusive snack definition, so the estimates are directly comparable to the raw-HEI heterogeneity of Section 4.3.1. Referenced from Section 4.3.1.

** $p < 0.05$, * $p < 0.10$.

Table B.4: Isocaloric snack substitution counterfactual.

<i>Panel A: substitution and whole-day score</i>							
Age bin	Snack kcal		Kcal	Share of	Whole-day HEI		
	N	per day	replaced	snack kcal	Before	After	Change
Ages 2–5	4,864	470	237	50.4%	52.6	60.1	+7.56
Ages 6–11	6,619	516	318	61.7%	47.1	56.9	+9.80
Ages 12–19	7,590	625	373	59.6%	45.1	56.0	+10.86
All ages 2–19	19,073	547	319	58.3%	47.7	57.3	+9.65

<i>Panel B: contribution by HEI component</i>				
HEI component	Ages 2–5	Ages 6–11	Ages 12–19	All ages 2–19
<i>Adequacy components (higher intake scores higher)</i>				
Total fruit	+1.17	+1.83	+2.13	+1.78
Whole fruit	+1.72	+2.23	+2.66	+2.27
Total vegetables	−0.12	−0.12	−0.13	−0.12
Greens and beans	0.00	0.00	0.00	0.00
Whole grains	+0.61	+0.68	+0.83	+0.72
Dairy	+1.15	+1.58	+1.89	+1.59
Total protein foods	+0.19	+0.20	+0.18	+0.19
Seafood and plant protein	+0.71	+0.78	+0.78	+0.77
Fatty acids	−0.64	−0.56	−0.55	−0.57
<i>Moderation components (lower intake scores higher)</i>				
Refined grains	+1.23	+1.18	+0.99	+1.12
Sodium	+0.03	−0.08	−0.16	−0.08
Added sugars	+1.79	+2.26	+2.53	+2.25
Saturated fats	−0.28	−0.18	−0.30	−0.26
Total	+7.56	+9.80	+10.86	+9.65

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, HEI sample, snacking children only, author’s calculations.

Notes: Sweet and salty snack items are removed from each child’s day and the same number of calories is added back as a basket of foods the children in this sample already consume at snack occasions. Items removed are cookies and cakes, salty grain snacks, candy and sweets, ice cream and frozen desserts, sugar-sweetened beverages, sports and energy drinks, and potato chips. The replacement basket is 40 percent fresh whole fruit, 20 percent plain yogurt, 15 percent natural cheese, 10 percent unsalted nuts and seeds, 10 percent air-popped popcorn, and 5 percent whole-grain crackers. Total daily calories are held fixed and the whole-day HEI-2015 is rescored per child. Panel B reports each component’s contribution to the change in Panel A, signed so that a positive value is an improvement in the score. This is an accounting exercise rather than a causal estimate. It assumes full compliance and no behavioral response, so the change in Panel A is best read as an envelope on what a composition change could deliver.

Meal-piece outcome	Tail-symmetric (% chg)	Normal (% chg)	Mean (per day)
<i>Panel A: adequacy components (higher-is-better on the HEI)</i>			
Fruit (cup eq)	+0.92 (0.76)	+1.63 (1.17)	0.62
Vegetables (cup eq)	−0.42 (0.62)	−0.43 (0.89)	0.78
Whole grain (oz eq)	−0.87 (1.16)	−0.67 (1.58)	0.49
<i>Panel B: moderation components (lower-is-better on the HEI)</i>			
Refined grain (oz eq)	+1.31** (0.47)	+2.16** (0.65)	4.80
Added sugar (g)	+4.09** (0.62)	+6.26** (0.89)	10.94
Saturated fat (g)	+1.98** (0.46)	+2.98** (0.63)	19.01

Table B.5: Bunching estimates of the effect of snack caloric intake on six per-child intake outcomes measured on the meal piece of the day, expressed as the percent change in meal-piece intake per 100 kcal of habitual snack intake.

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, HEI sample, $N = 21,508$, $K=50$, $\text{boot_reps}=30$, author’s calculations.

Notes: Each row reports the bunching coefficient of a meal-piece intake outcome on `snack_kcal_1`, scaled as $100 \times (100 \times \hat{\beta})/\bar{Y}$ so that a value of +1 reads as a one-percent increase in meal-piece intake per 100 kcal of habitual snack intake. Meal-piece totals sum over the individual food file rows where the eating occasion is breakfast, lunch, or dinner, or where a calorie-bearing drink was reported within 30 minutes of one of those meals. Standard errors in parentheses are bootstrap standard errors based on 30 replications. The Tail-symmetric column is the nonparametric tail-symmetric estimator used as the main-body headline; the Normal column is the semiparametric-normal alternative. Referenced from Section 4.4.

** $p < 0.05$, * $p < 0.10$.

Table B.6: Federal child-nutrition policies and where the snacking margin fits, by target margin, indicative federal cost, and available causal-effect estimate.

Policy	Target margin	Federal cost (recent yr)	Documented effect on child health
NSLP + SBP	Composition of school-provided meal-occasion food, students of eligible schools	~\$18B (NSLP) + ~\$5B (SBP)	Under the 2010 Healthy, Hunger-Free Kids Act (HHFKA) nutrition standards, phased in from 2012, the whole-day HEI of NSLP participants improved by 1.9–4.7 points depending on income tercile in a difference-in-differences design comparing pre- and post-HHFKA cohorts, with the larger gains at lower income (Kinderknecht et al., 2020). Effects on child BMI mixed and small, with earlier evidence indicating a small positive association between school-lunch participation and BMI (Hinrichs, 2010; Schanzenbach, 2009)
SSB taxes (Berkeley, Philadelphia, Seattle, others)	Retail price of taxed beverages; consumer substitution	Near-zero fiscal cost (revenue-raising)	Reduces taxed-beverage purchases by 20–40% in adopting municipalities. Total sugar and calorie intake not cleanly reduced because of consumer substitution to untaxed drinks and cross-border shopping. No consistent effect on child BMI documented at the population scale (Allcott et al., 2019; Cawley & Frisvold, 2017)
SNAP	Household food-purchase budget for eligible low-income households	~\$120B	Substantial improvements in household food security. Effects on child diet quality are small and heterogeneous. Participation-obesity relationship confounded by selection, with bounded estimates covering zero (H. W. Hoynes & Schanzenbach, 2009; Kreider et al., 2012). No direct restriction on food composition or eating occasion
<i>Snacking margin (this paper)</i> candidate levers: parental information, retail labeling, market-ing restrictions	Habitual child snack intake outside structured meals	No existing federal program	Bunching estimate: -0.52 HEI points per 100 kcal of habitual snacking (pooled), -1.12 points at ages 2–5; the effect covers the full day rather than a meal-piece slice or a single food category. Homogeneous across household-income terciles. BMI z-score effect a precise null (upper bound 0.019 per 100 kcal at the 5% level)

Data source: Federal-cost figures from USDA Food and Nutrition Service program-summary reports (most recent available fiscal year); causal-effect summaries from the cited papers.

Notes: The table is a compact policy-comparison view of where the paper’s estimated snacking margin sits relative to the three most-studied federal child-nutrition instruments. Federal-cost numbers are indicative and drawn from the most recent USDA program-summary reports at the time of writing; they are not internally comparable in a cost-per-unit-of-HEI-improvement sense because the four policies target different populations and different margins. Referenced from Section 6.

C Figures



Figure C.1: Mean HEI-2015 component scores for the full-day diet, split by age group (left panel, three age bins) and by household income tercile (right panel, defined by the poverty income ratio).

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, HEI sample, author’s calculations.

Notes: Bars show the mean score on each HEI-2015 component by group, normalized to a 0–100 scale (% of the component’s maximum) and ordered by the overall mean. Groups are shaded light to dark, from youngest to oldest in the left panel and from lowest to highest income in the right panel. Reverse-coded components (added sugars, sodium, refined grains, saturated fats) are oriented so that a higher score always indicates a better diet.

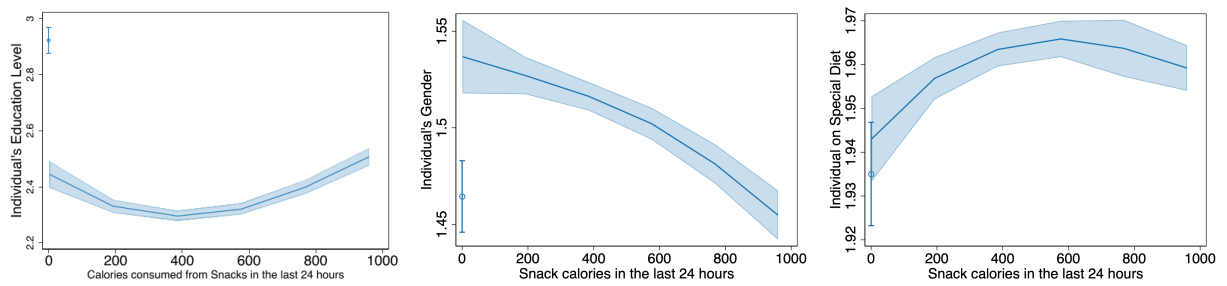


Figure C.2: Local mean as a function of snack caloric intake S_i in a neighborhood of zero, for the remaining controls in X_i not shown in Figure 3 and for the child's dieting status.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018, author's calculations.

Notes: Left to right, the panels show the child's educational attainment, an indicator for the child's gender, and an indicator for whether the child reports being on a special diet. The dieting indicator is not part of X_i and is shown as an additional balance check. The point at $S_i = 0$ is the mean among bunched children with a 95% confidence interval, and the line on the positive side is a degree-1 local polynomial smoother on snackers with a bandwidth of 250 kcal.

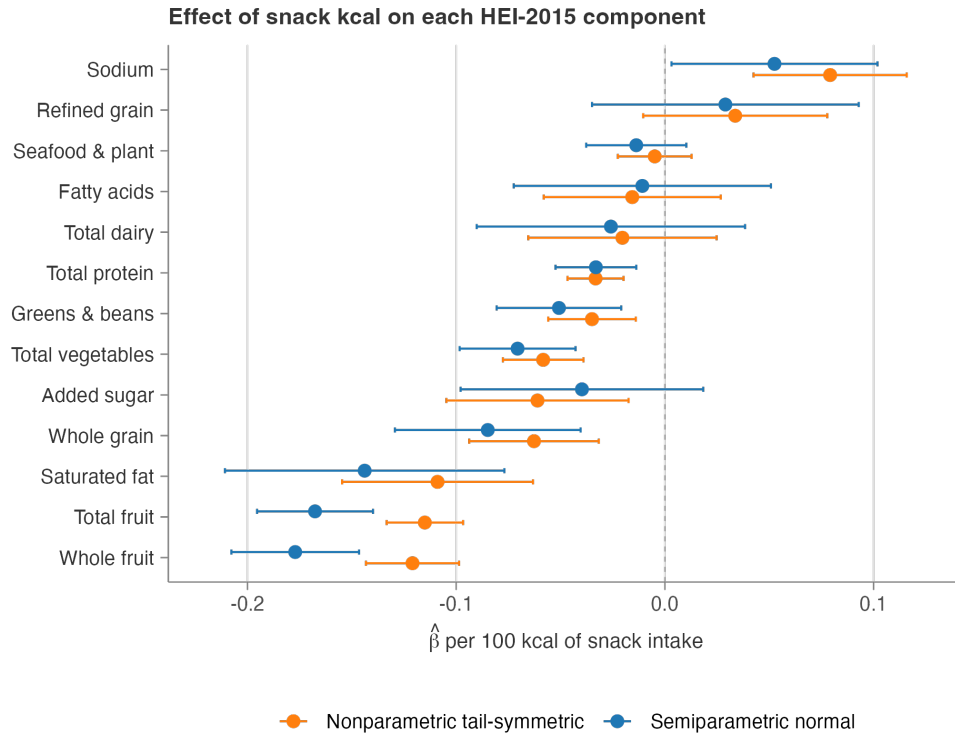


Figure C.3: Bunching estimates of the effect of snack caloric intake on each of the 13 HEI-2015 components.

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, HEI sample, author’s calculations.

Notes: Each row is one HEI-2015 component, ordered by the tail-symmetric point estimate with the largest declines at the bottom. The coefficient is the per-100-kcal effect of snack intake on that component’s score, on the raw HEI-2015 component scale (component maxima of 5 or 10), so the thirteen component effects sum to the effect on the total score. Dots are point estimates with 95% bootstrap confidence intervals, in orange for the nonparametric tail-symmetric estimator and blue for the semiparametric-normal estimator. Vertical dashed line at zero. Referenced from Section 4.1.

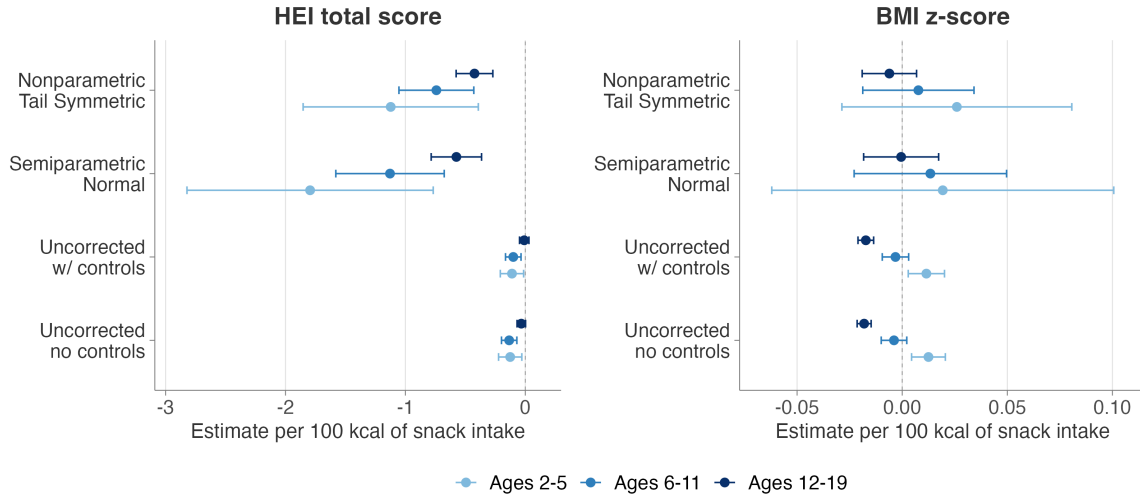


Figure C.4: Bunching estimates of the effect of snack caloric intake by age group.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI panel) and 2005–2018 (HEI panel), author’s calculations.

Notes: The left panel reports the HEI total score and the right panel the BMI z-score, each on its own horizontal scale. Within a panel the four rows are the specifications of Figure 7, namely the two uncorrected estimators and the two bunching-corrected estimators, with the preferred nonparametric tail-symmetric estimator in the top row. The three colored series are the age groups, shaded from light for the youngest to dark for the oldest. Each point is the coefficient per 100 kcal of snack intake with its 95% confidence interval as a horizontal error bar. Vertical dashed line at zero. Referenced from Section 4.3.1.

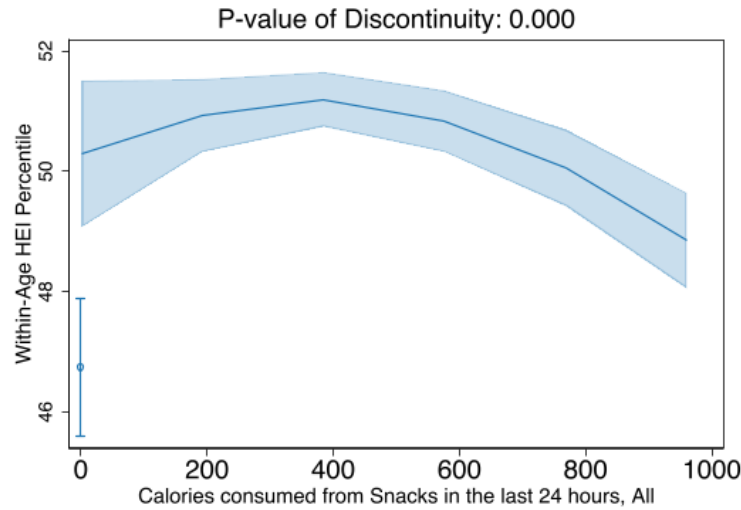


Figure C.5: Raw outcome discontinuity at the bunching point for the within-age HEI percentile rank.

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, author’s calculations.

Notes: The point at $S_i = 0$ is the mean among bunched children with a 95% confidence interval, and the line on the positive side is a degree-1 local polynomial smoother on snackers with a bandwidth of 250 kcal. Referenced from Section 4.3.1.

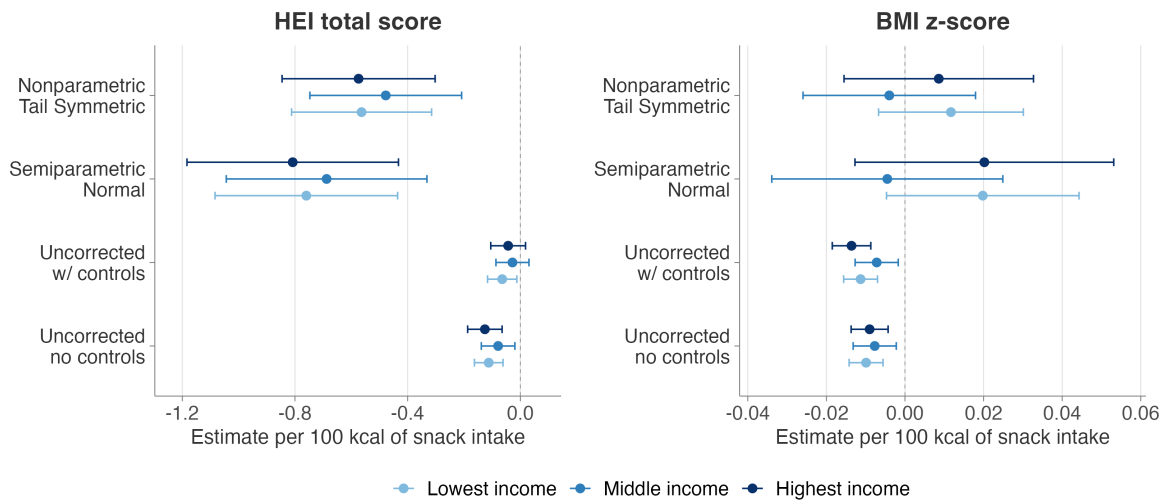


Figure C.6: Bunching estimates of the effect of snack caloric intake by household income tercile.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI panel) and 2005–2018 (HEI panel), author’s calculations.

Notes: The left panel reports the HEI total score and the right panel the BMI z-score, each on its own horizontal scale. Within a panel the four rows are the specifications of Figure 7, namely the two uncorrected estimators and the two bunching-corrected estimators, with the preferred nonparametric tail-symmetric estimator in the top row. The three colored series are the household income terciles, shaded from light for the lowest to dark for the highest. Each point is the coefficient per 100 kcal of snack intake with its 95% confidence interval as a horizontal error bar. Vertical dashed line at zero. Referenced from Section 4.3.2.

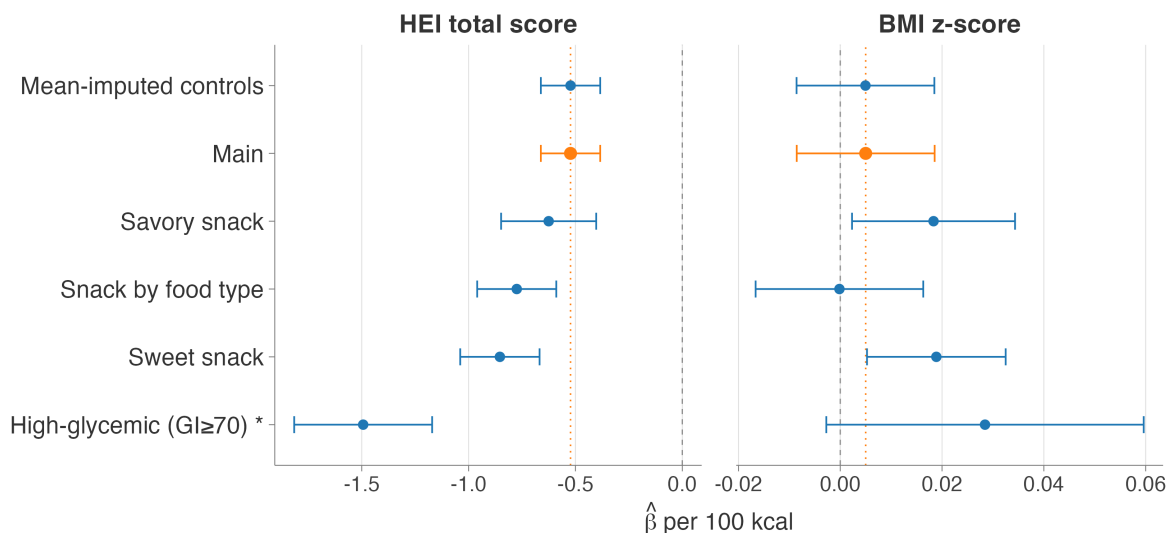


Figure C.7: Specification curve of the bunching estimate across the snack definitions discussed in the text, on both headline outcomes.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI sample) and 2005–2018 (HEI sample), $K = 50$, author’s calculations.

Notes: The left panel reports the HEI total score and the right panel the BMI z-score, each on its own horizontal scale. Each row is one definition of the snack treatment, plotted as the per-100-kcal point estimate with its 95% bootstrap confidence interval and ordered by the HEI effect. The main drinks-inclusive treatment is in orange, with a dotted line marking its estimate in each panel. The remaining rows are the sweet and savory composition split, the high-GI treatment (glycemic index at or above 70), the alternative food-type snack definition, and a variant that recodes missing controls to the sample mean rather than to zero. All rows use the preferred nonparametric tail-symmetric estimator except the high-GI treatment (marked *), whose 83% bunching share rules out tail symmetry, so its semiparametric-normal estimate is reported instead (its construction is documented in Appendix A.1). Referenced from Section 4.3.3 and Section 5.4.

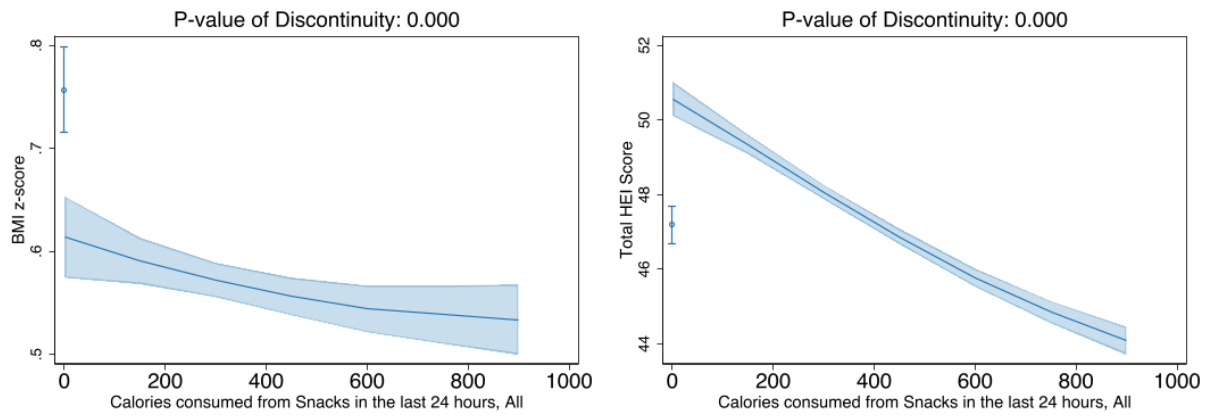


Figure C.8: Raw outcome discontinuity at the bunching point under the food-classification snack definition.

Data source: National Health and Nutrition Examination Survey (NHANES), 2003–2018 (BMI panel) and 2005–2018 (HEI panel), author’s calculations.

Notes: Left panel: BMI z-score against S_i in a neighborhood of zero. Right panel: HEI total score. The point at $S_i = 0$ is the mean among bunched children with a 95% confidence interval, and the line on the positive side is a degree-1 local polynomial smoother of the outcome on snackers ($S_i > 0$) with a bandwidth of 250 kcal. Bunching share under this definition is 12.6% in the BMI sample. Referenced from Section 5.4.



Figure C.9: Mean HEI-2015 component scores among snackers, comparing the full-day diet, the snack-only diet, and the breakfast-only diet.

Data source: National Health and Nutrition Examination Survey (NHANES), 2005–2018, author’s calculations.

Notes: The snack-only diet is the HEI computed using only the foods consumed during snack occasions, and the breakfast-only diet is the HEI computed using only the foods consumed at breakfast occasions, included as a reference for an eating occasion that is rarely a snack event. On average the snack-only score totals 42.5, the breakfast-only score totals 43.4, and the full-day score totals 50.0 for the same children. Each component is normalized to a 0–100 scale (% of its HEI-2015 maximum) so that the 13 axes (whose raw maxima are either 5 or 10) share a common scale. Reverse-coded components (added sugars, sodium, saturated fats, refined grains) are oriented so that a higher score always indicates a better diet.